

Digitopia Phase-III | Undergraduate Students | Disabilities Challenge

FEEGNet: Bridging Thought to Action

An Advanced Brain-Computer Interface for
Enhanced Human Communication and Universal
Accessibility

Final Presentation | 29th SEP 2025 | Team NeuroCon

Meet the team



Mahmoud Ezzat

Edge AI Enthusiast & AI
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Senior Computer Engineering
Student



Marina Mohsen

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Senior Computer Engineering
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Our Roles



Mahmoud Ezzat

- Designed & trained FEEGNet MI architecture
- Designed MI Signal Processing Pipeline
- Ensured 82% Accuracy in MI Tasks

Our Roles



Marina Mohsen

- Designed & trained ensemble ML SSVEP architecture
- Designed SSVEP Signal Processing Pipeline
- Ensured 92% Accuracy in SSVEP Tasks

Our Roles



Youssef Badr

- Developed adaptive spatial attention & physiological feature extraction
- Built real-time inference system with 0.34 RTF
- Led simulation and optimization

The Silent Tragedy

"Imagine being unable to say 'I am hungry,'
'I am in pain,' 'thank you,' or 'I love you.'"

Being trapped inside your body - a body that doesn't
respond to commands.

Surrounded by people, yet utterly alone."

Martin Pistorius, Locked-in Syndrome Survivor (13 years)

The Shocking Reality

963

per 100,000 Egyptians
live with stroke consequences

1 Million+

Egyptians today
living with communication barriers

210,000

new strokes yearly
One every 2.5 minutes

5.4 Million

with motor disabilities
Unable to use standard technology

The Human Cost

Every single day in Egypt: **575 people** have strokes, 40 become permanently disabled, **15 develop locked-in syndrome**. This is our national emergency.



The Silent Tragedy

“I was aware of everything, just like any normal person, I could see and understand everything, but I couldn't find a way to let anybody know, my personality was entombed within a seemingly silent body, A vibrant mind hidden in plain sight “

Martin Pistorius, Locked-in Syndrome Survivor (13 years)

A DAY IN SILENCE

Wakes up hungry → **can't say** "I am hungry."

Family asks if he's in pain → **cannot answer.**

Has an idea → **cannot share it.**

Wants to say "*thank you*" → **words locked inside.**

Needs urgent help → **no way to call.**



The Ripple Effect

- Families torn apart by daily caregiving burden
- Young professionals forced out of workforce
- Children grow up without hearing parents' voices
- Millions isolated from society, productivity, and dignity



But What If...

- What if **silence** could be broken?
- What if thought **alone** became a voice?
- What if every Egyptian **trapped in silence** could reconnect with life?



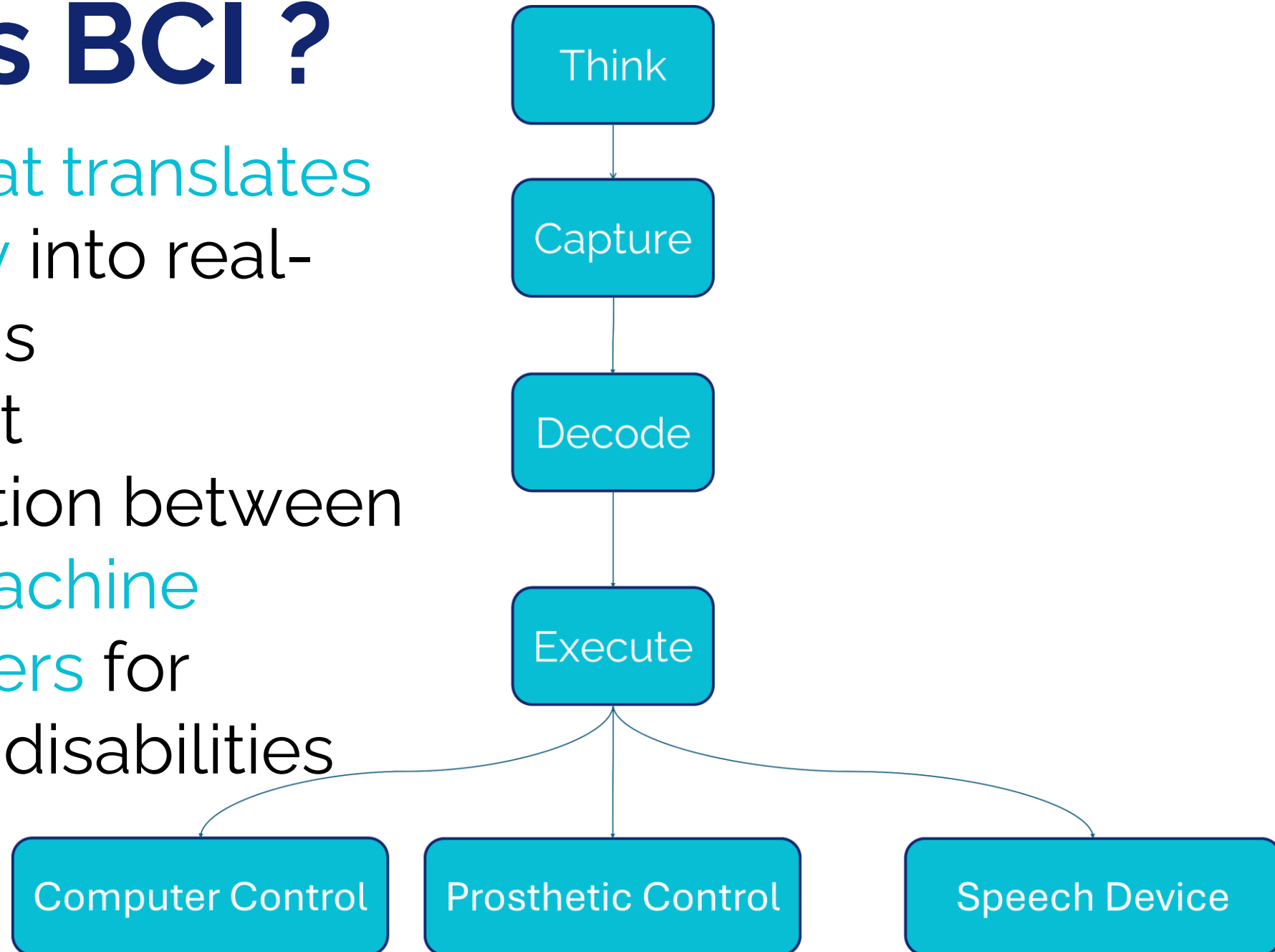
Finding Solution

- Translates thoughts directly into actions
- Restores communication, independence, and dignity



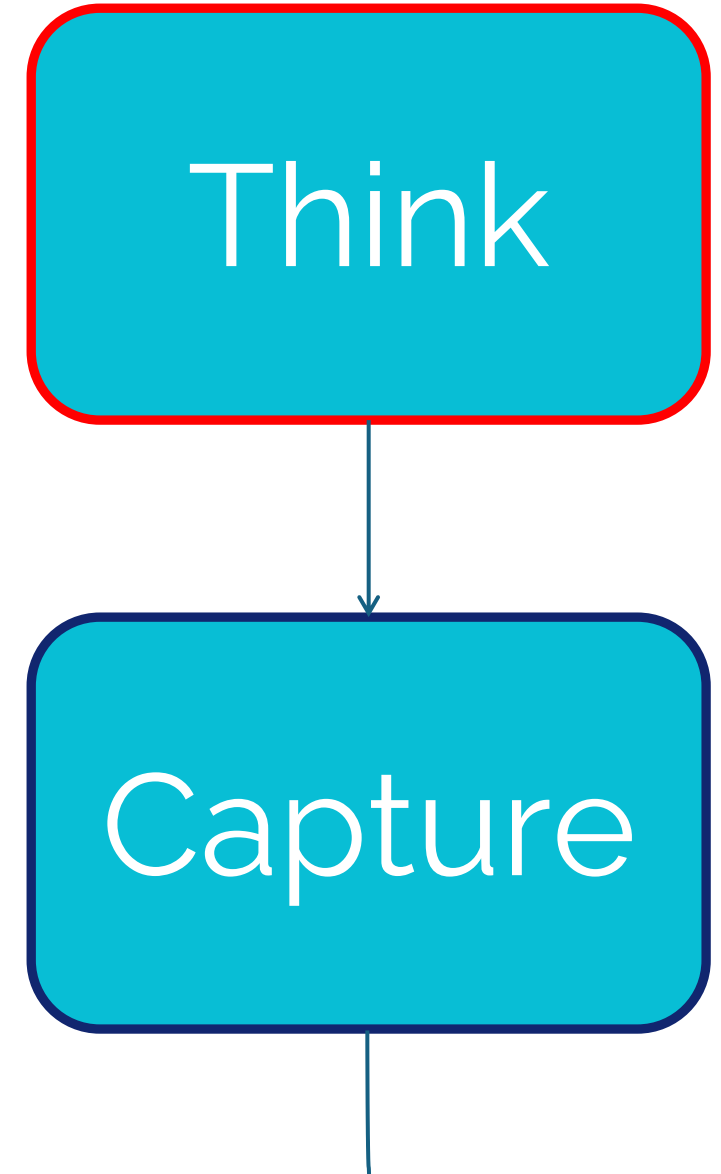
What is BCI ?

- A system that translates brain activity into real-world actions
- Allows direct communication between mind and machine
- Breaks barriers for people with disabilities

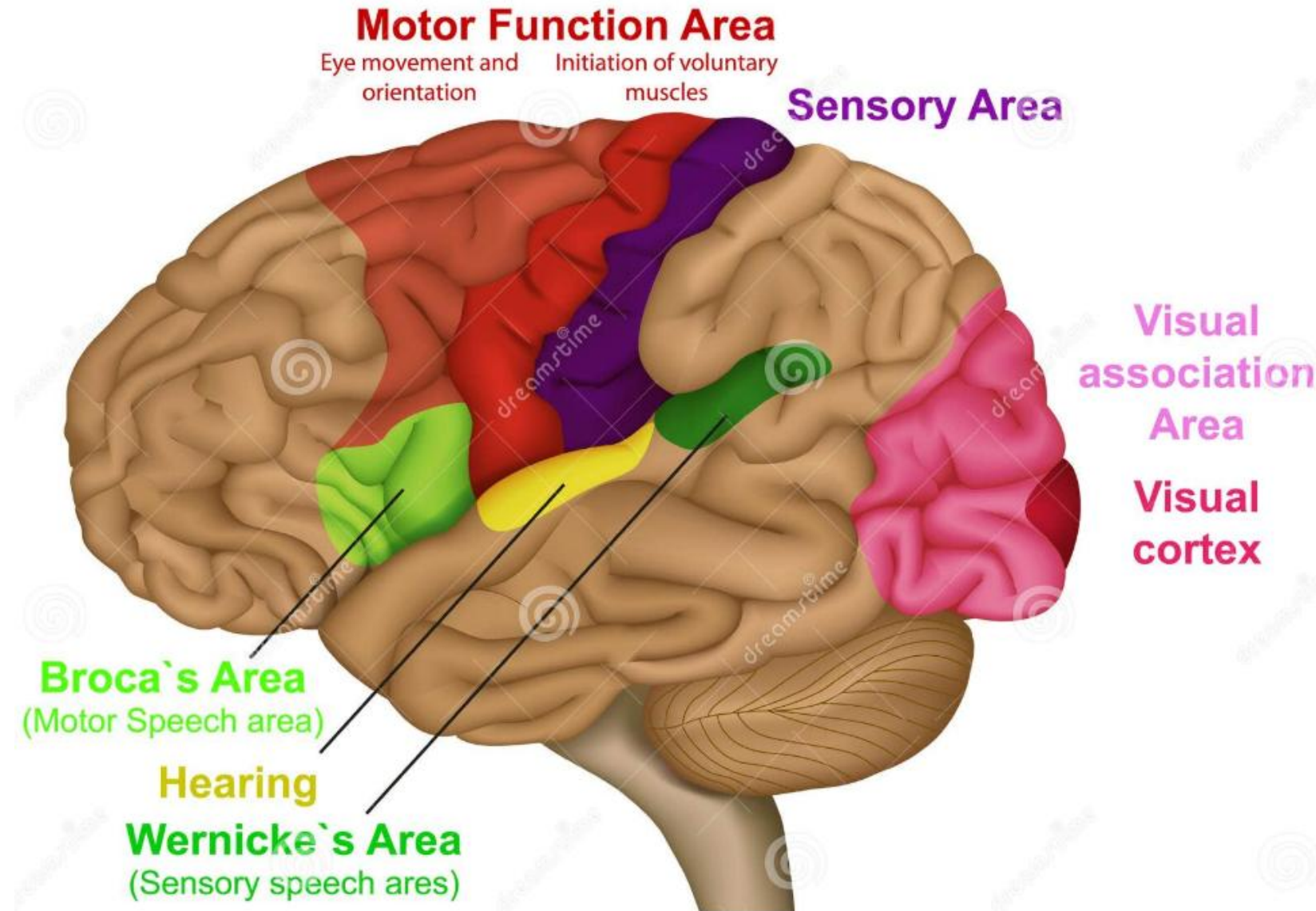


What is BCI ?

- User **imagines** a specific movement or focuses on a stimulus
- Example: imagine moving the right hand, or focus on flashing light
- Brain generates **unique electrical activity patterns**



What is BCI ?

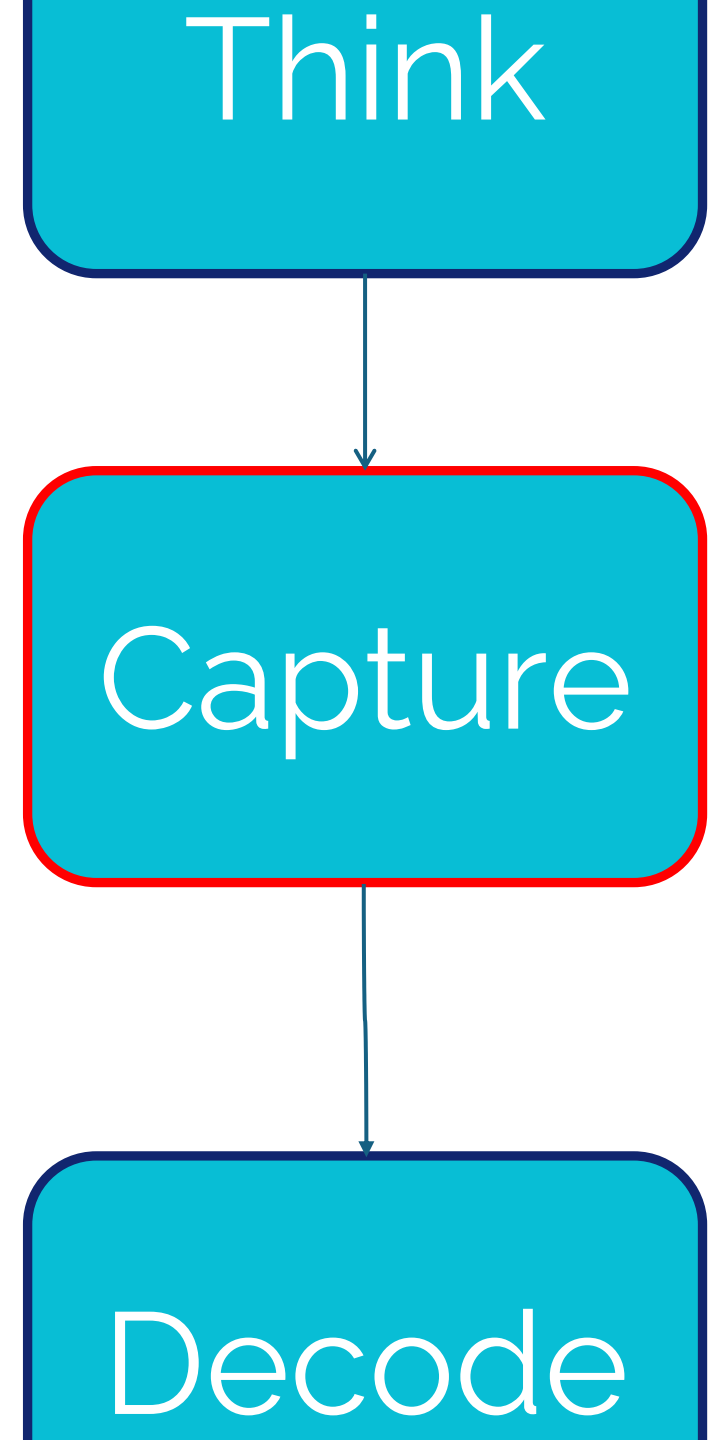


Think

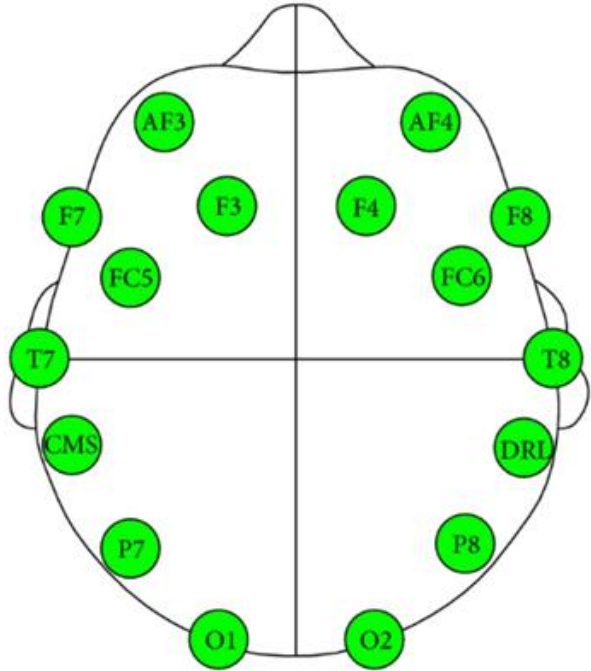
Capture

What is BCI ?

- Non-invasive headset with electrodes placed on scalp
- Brain activity measured using EEG (Electroencephalography)
- Detects millisecond-level electrical signals from neurons



What is BCI ?



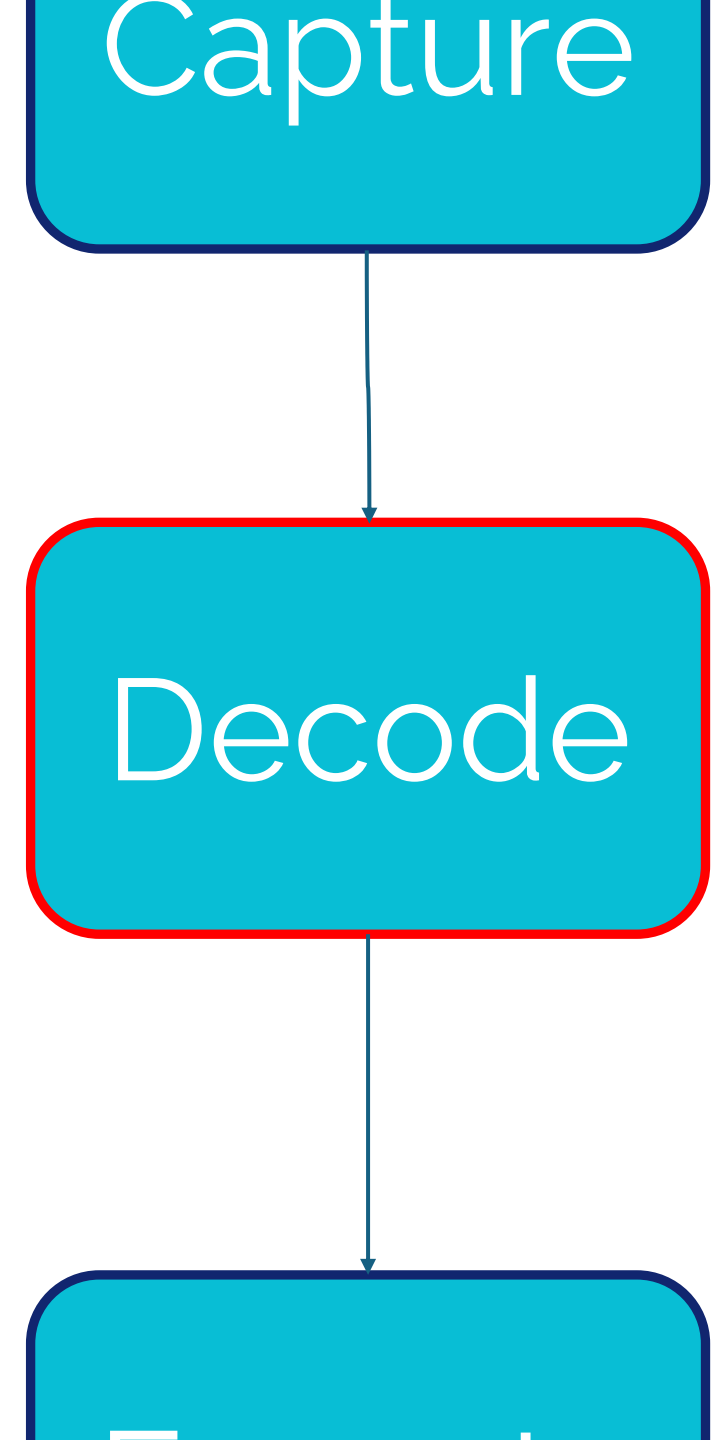
Think

Capture

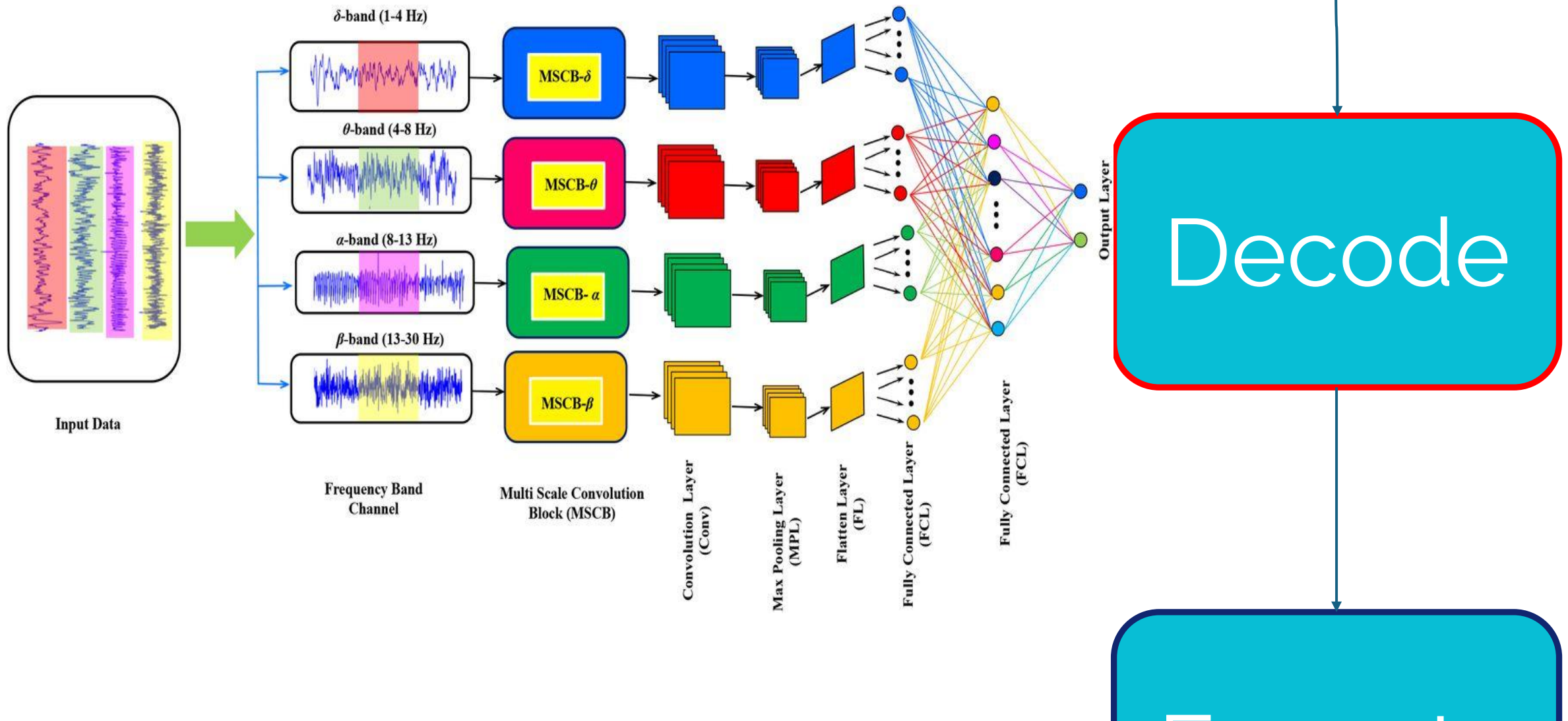
Decode

What is BCI ?

- AI interprets the brainwave patterns
- Machine learning identifies which thought corresponds to which command
- Example: right-hand imagery → “cursor right”



What is BCI ?

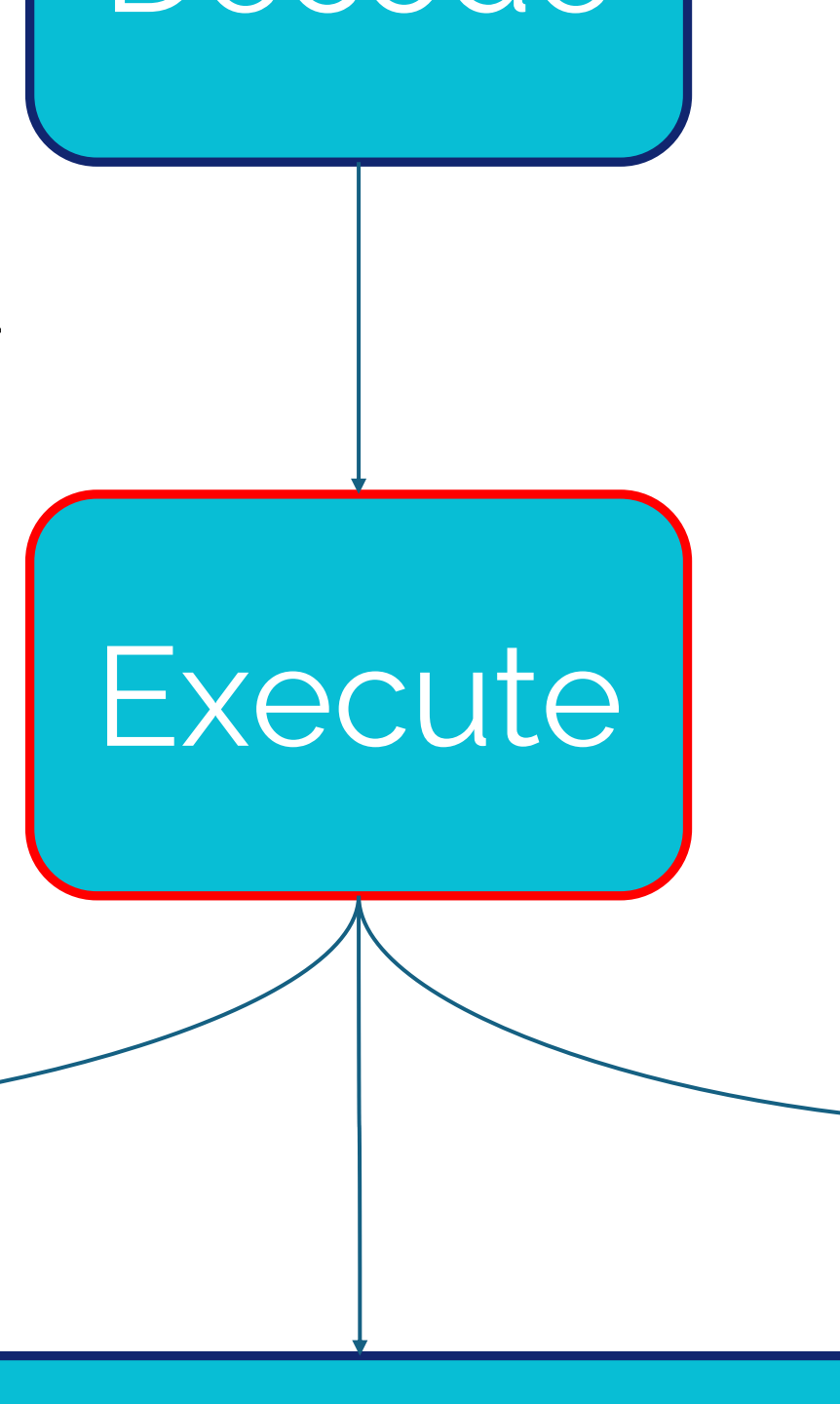


What is BCI ?

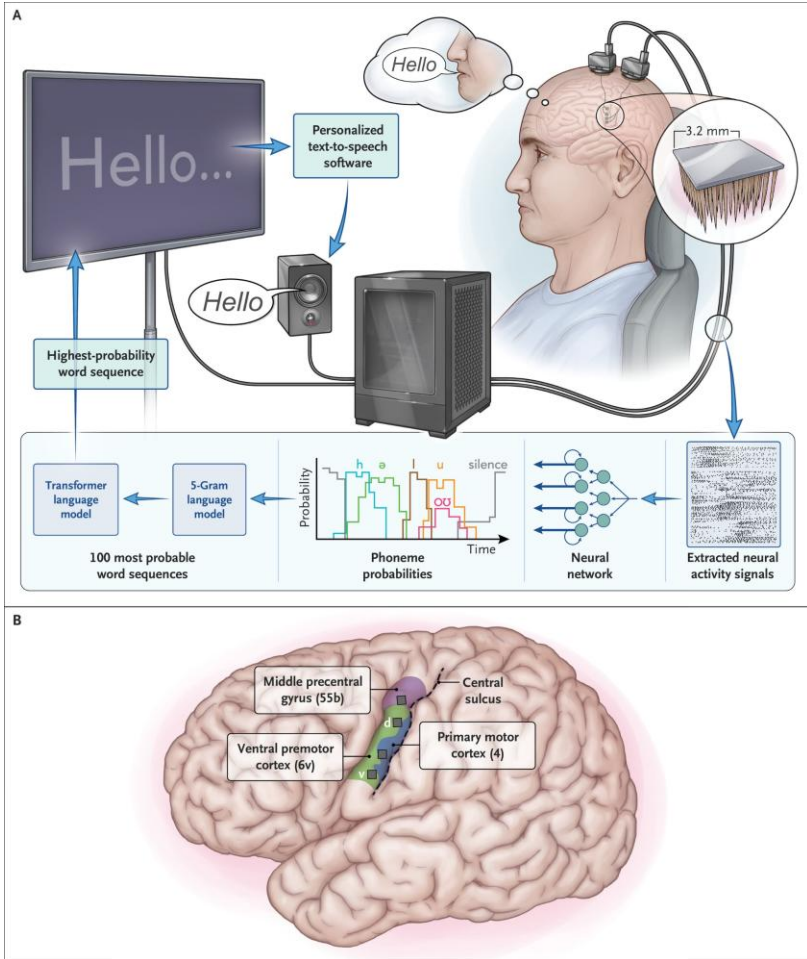
Decoded command triggers a real-world action

Applications:

- **Motor Control** – wheelchair steering, smart home control
- **Prosthetic Control** – moving artificial limbs
- **Speech Devices** – text-to-speech communication



What is BCI ?



Decide

Execute

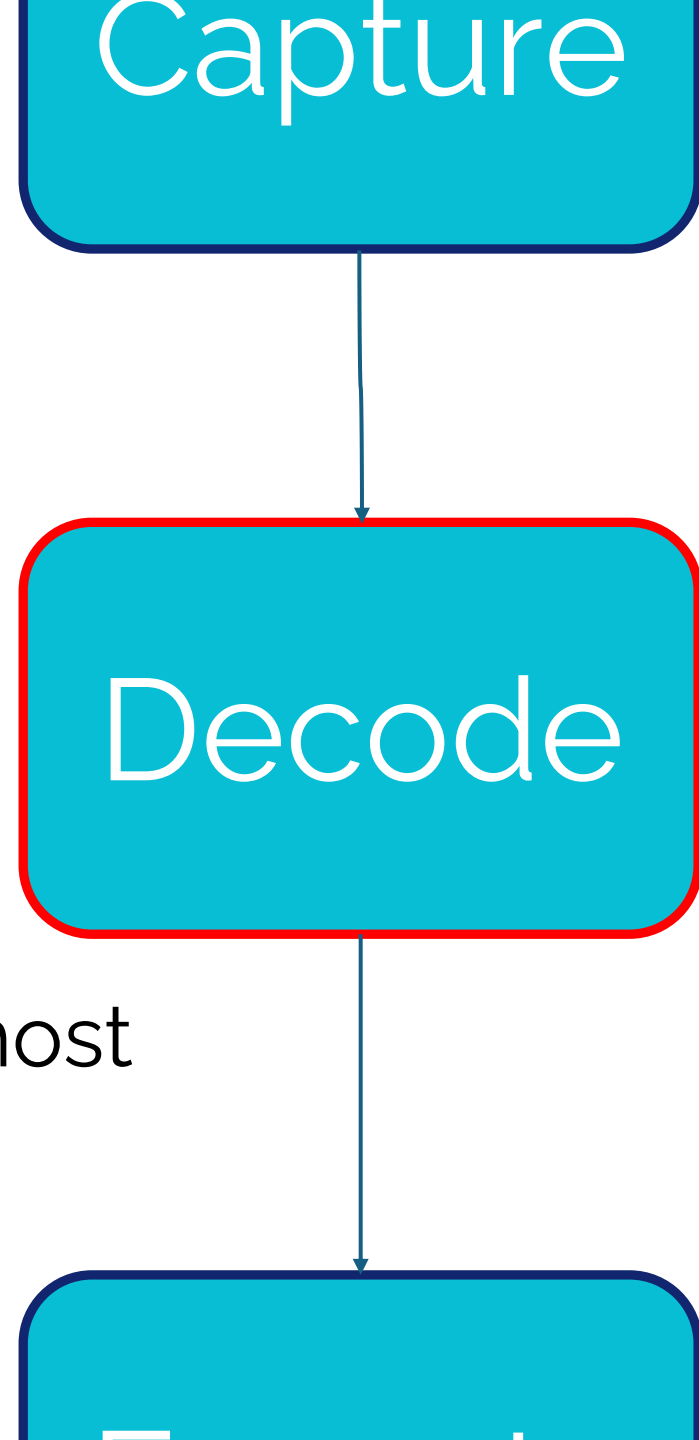
Seeing Opportunity

The Promise:

- Freedom through thought alone
- Seamless human-machine connection
- Independence restored

The Reality:

- Systems stuck in research labs
- Too complex, too expensive, too slow
- Out of reach for those who need them most



Why Previous BCIs Failed

- 128+ electrodes → Impractical, uncomfortable
- Hours of calibration → Frustrating setup
- 2–5 second delays → Unusable in real time
- \$50,000+ cost → Impossible for most families
- 50–70% accuracy → Unreliable, frustrating
- Needs expert technician → Not home-ready



What If We Could Fix All That?



What If We Could Fix All That?

Imagine a BCI that is:

- **Practical**: only a few electrodes
- **Real-time**: faster than a blink
- **Affordable**: accessible to Egyptian families
- **Accurate**: reliable enough for daily use

FEEGNet: The Egyptian Innovation that changes everything

Fast Efficient EEG Network

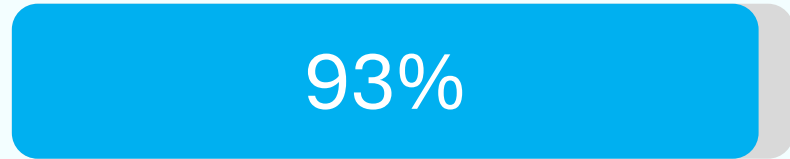
- **Fast Efficient EEG Network** – A practical, real-time BCI system
- **8 electrodes only** – comfortable, everyday use
- **< 0.34 RTF** – faster than a blink
- **Reliable** – 82–93% accuracy, works anywhere

What is FEEGNet

- **AI Model for Brain Decoding**
 - Learns to translate brainwave patterns into commands
 - Works with both **Motor Imagery (MI)** and **SSVEP** signals
- **Motor Imagery (MI)**
 - User imagines left/right hand movements
 - FEEGNet detects motor cortex activity patterns
- **Steady-State Visual Evoked Potential (SSVEP)**
 - User looks at flickering lights of specific frequencies
 - FEEGNet detects frequency-locked brain rhythms
- **End Result**
 - Thought → AI decoding → Action (cursor, wheelchair, speech device)

World-Record Performance

SSVEP Classification

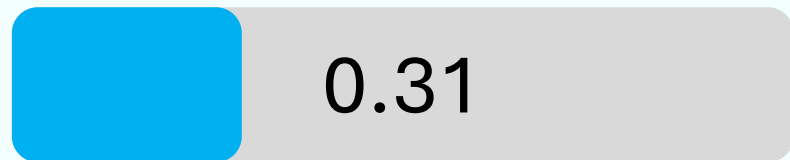


MI Classification

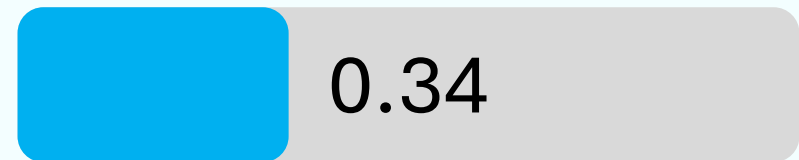


One of the world highest reported accuracy

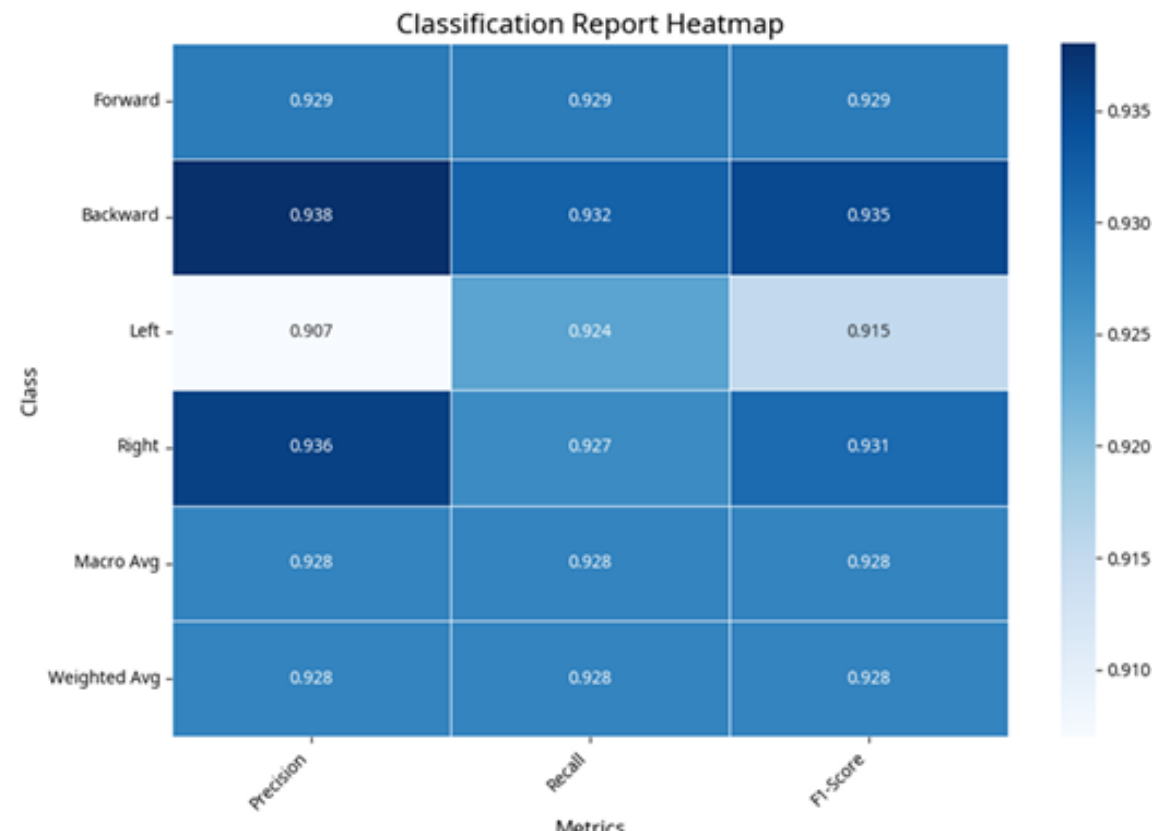
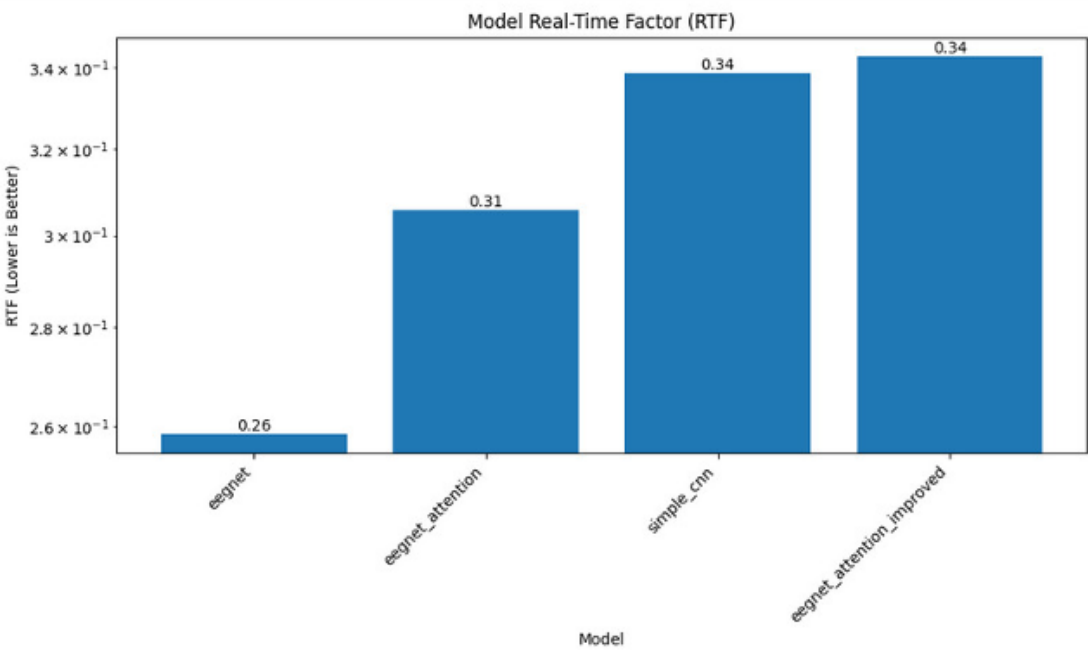
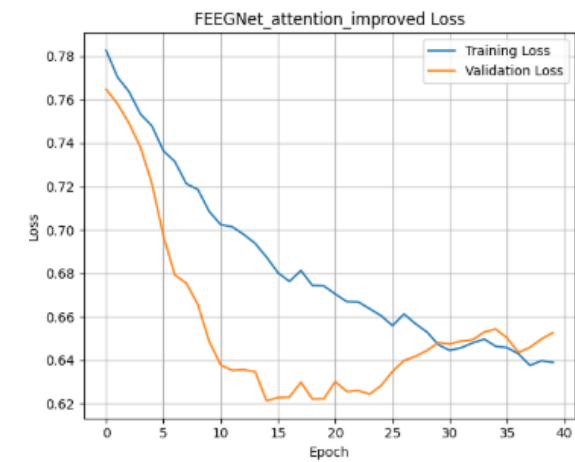
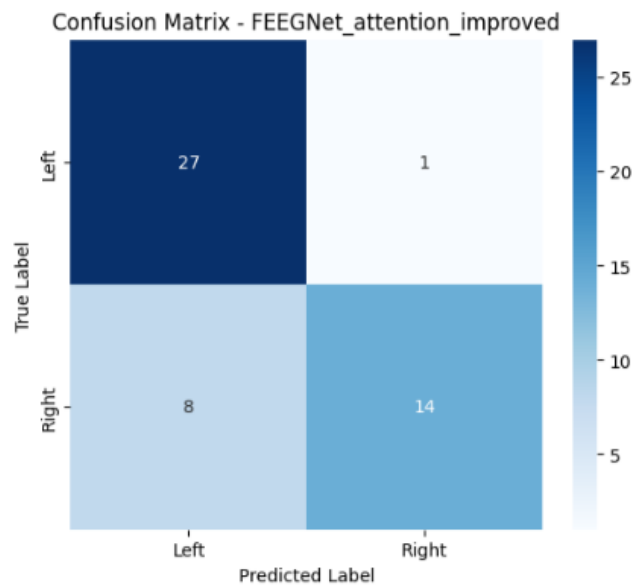
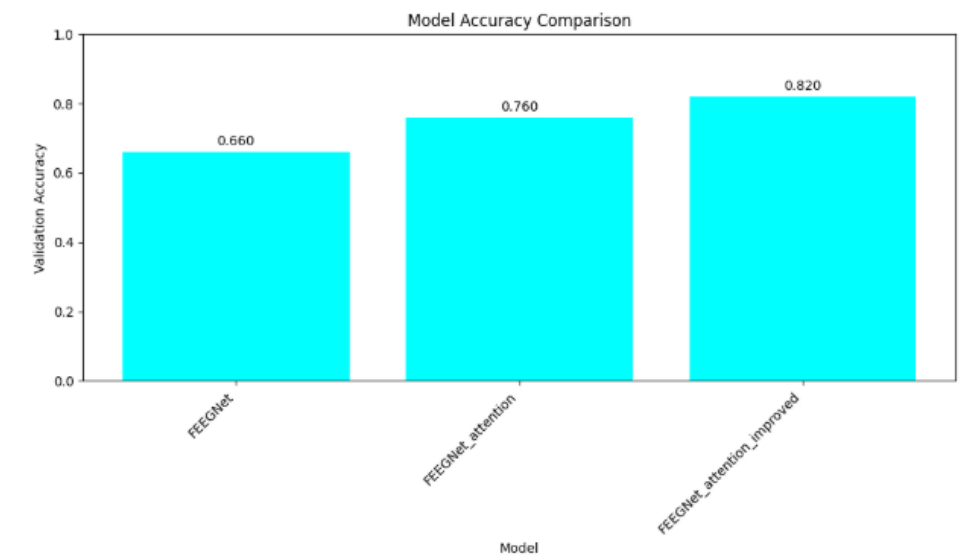
SSVEP Response



MI Response



At the speed of thought



Inside FEEG: Our Innovation

Adaptive Spatial Attention

- Learns which brain regions matter most
- Focuses on motor cortex (C3/C4) for hand movements
- Emphasizes occipital areas for visual tasks

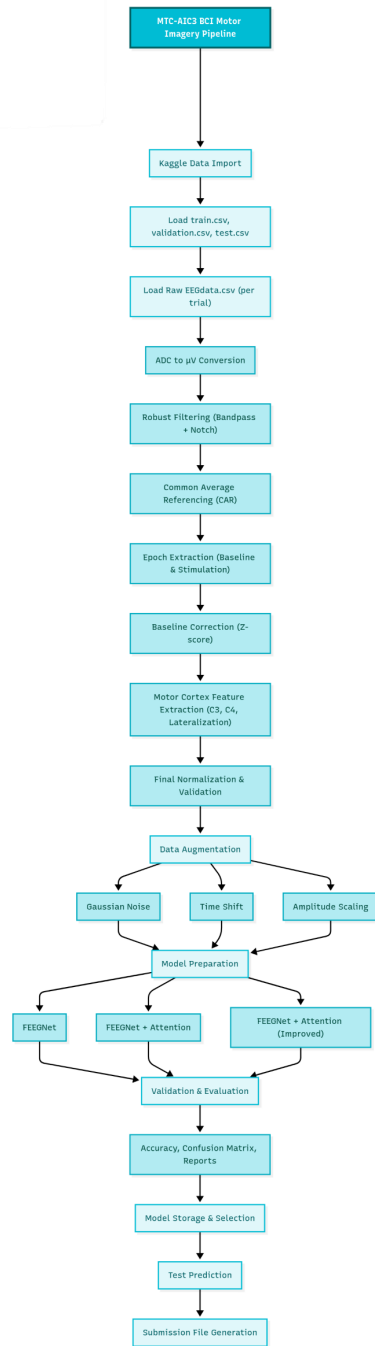
Physiological Feature Engineering

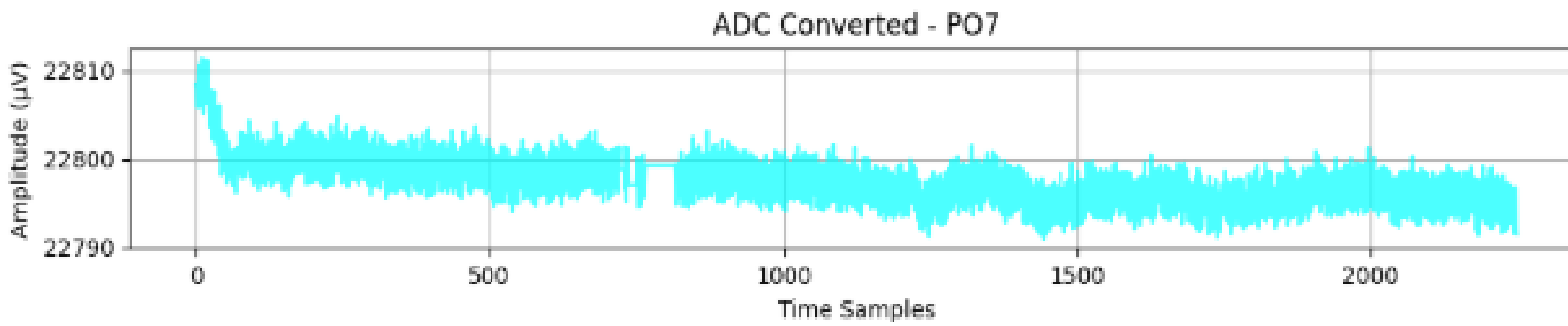
- Directly models brain's natural lateralization
- Embeds neuroscience into AI design

Universal Generalization

- Works across different users without retraining
- Smart data augmentation resists noise & artifacts

Technical Details



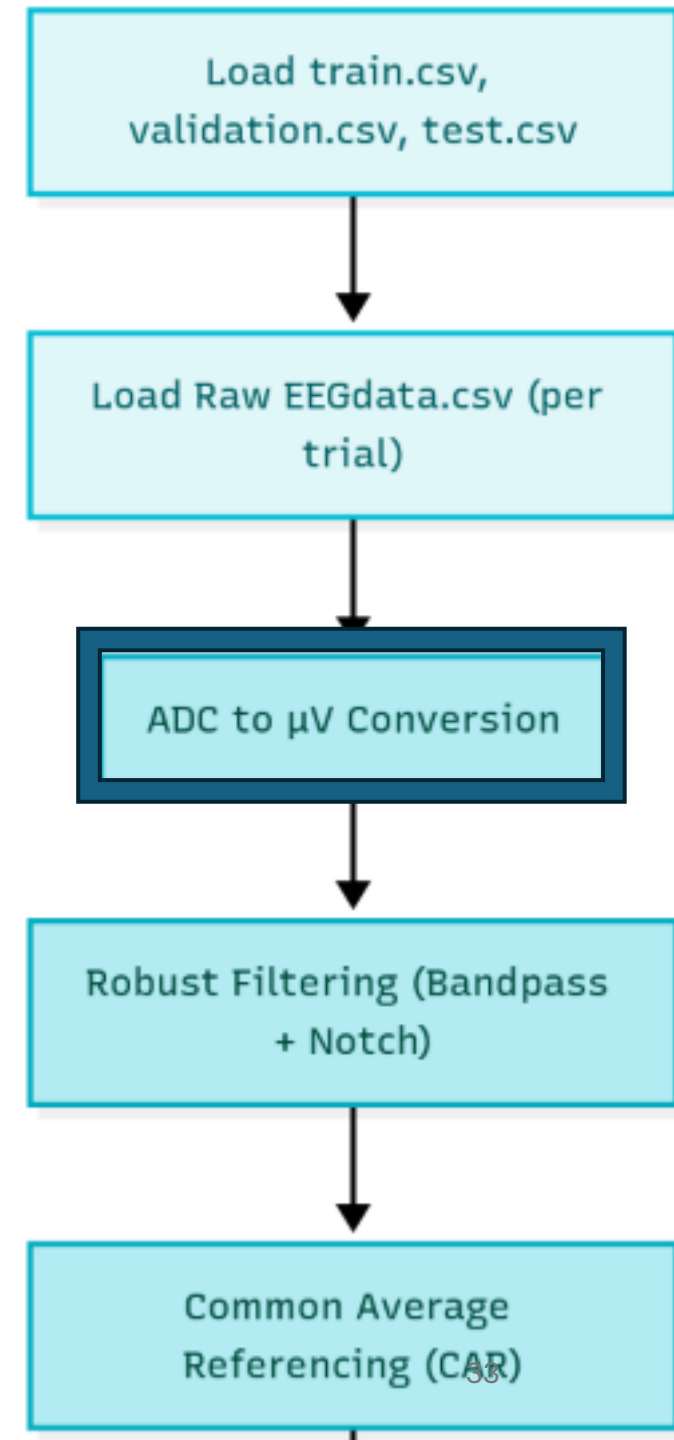


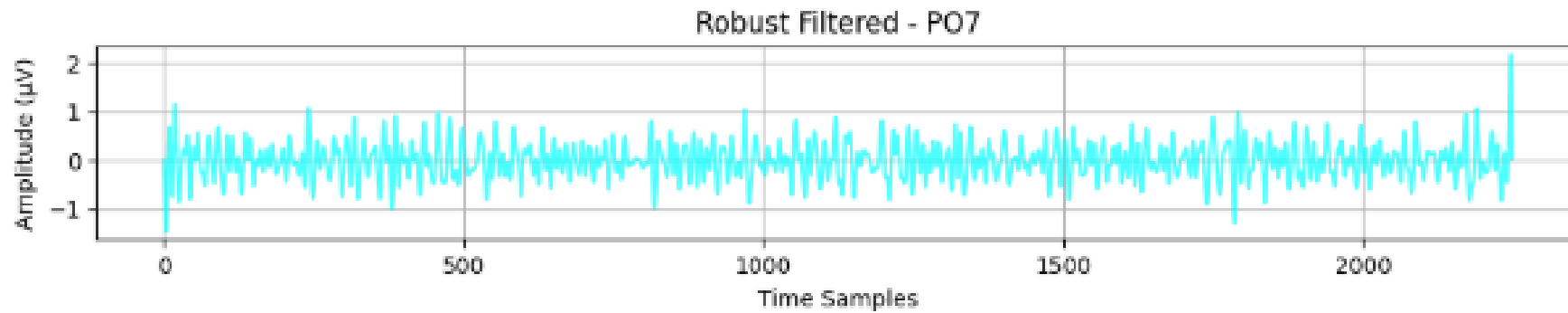
Objective

To convert the raw, arbitrary Analog-to-Digital Converter (ADC) units from the hardware into a standard, interpretable scientific unit: **microvolts (μV)**.

Benefit

standardizes the data, providing a clean and universally understood foundation for the entire processing pipeline.





Objective

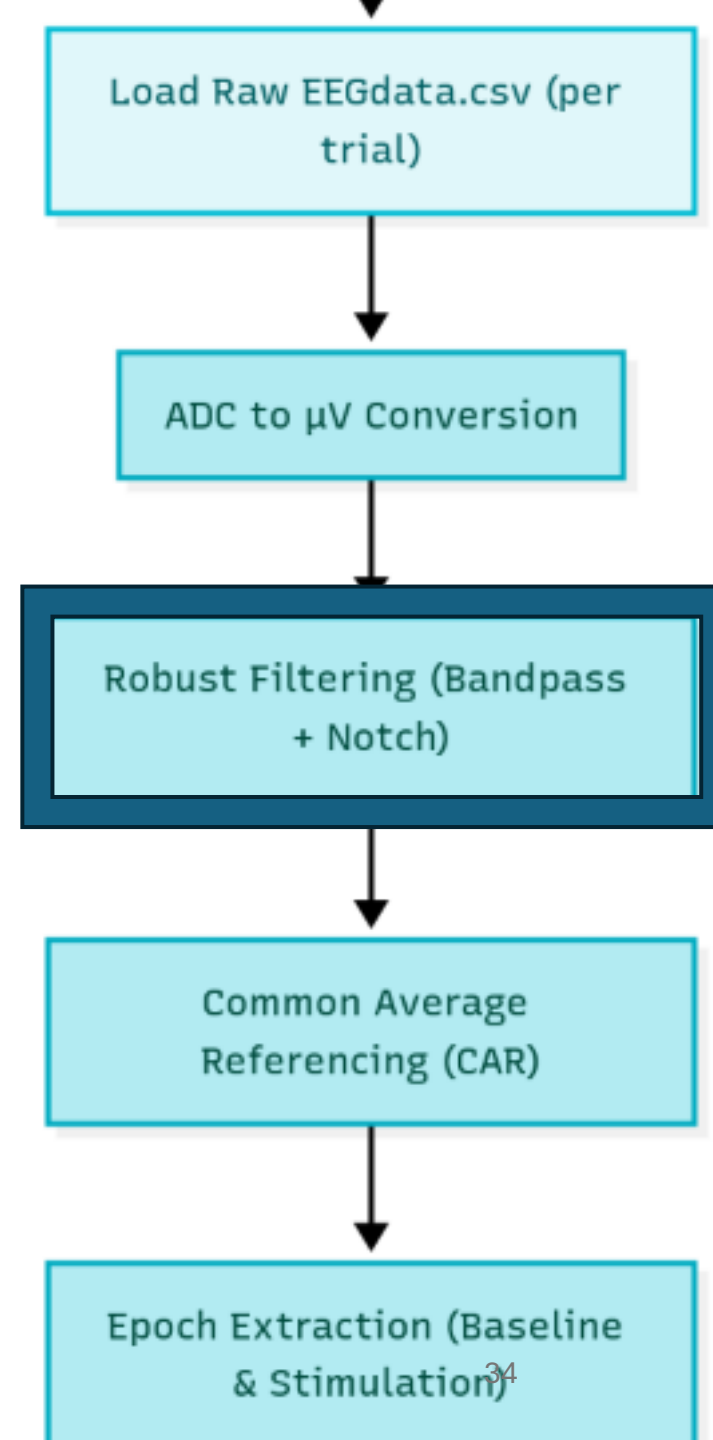
To isolate the neurologically relevant frequency bands for Motor Imagery and eliminate common sources of environmental and physiological noise from the µV-converted signal.

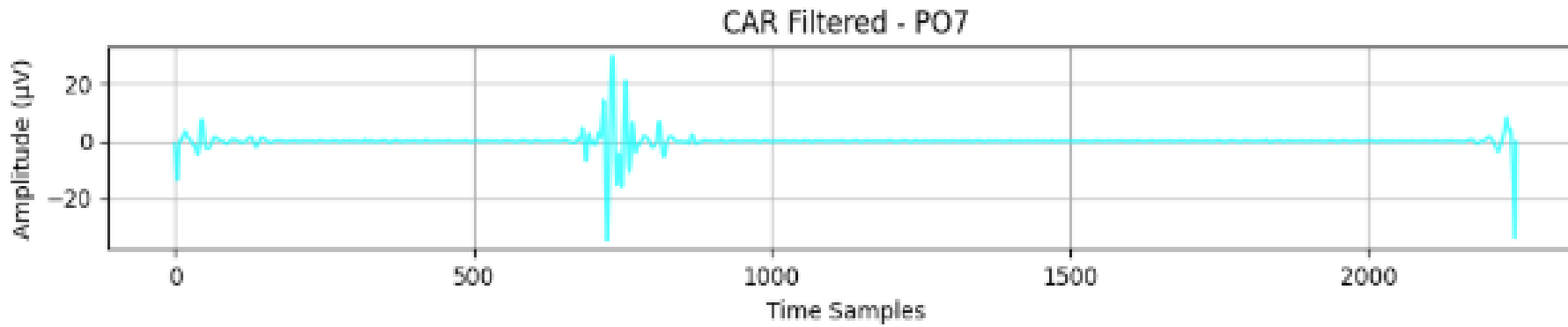
Benefit

yields a clean signal focused exclusively on the brain activity relevant to the MI task

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s}{\omega_c}\right)^{2n}}}$$

$$H(z) = \frac{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$$





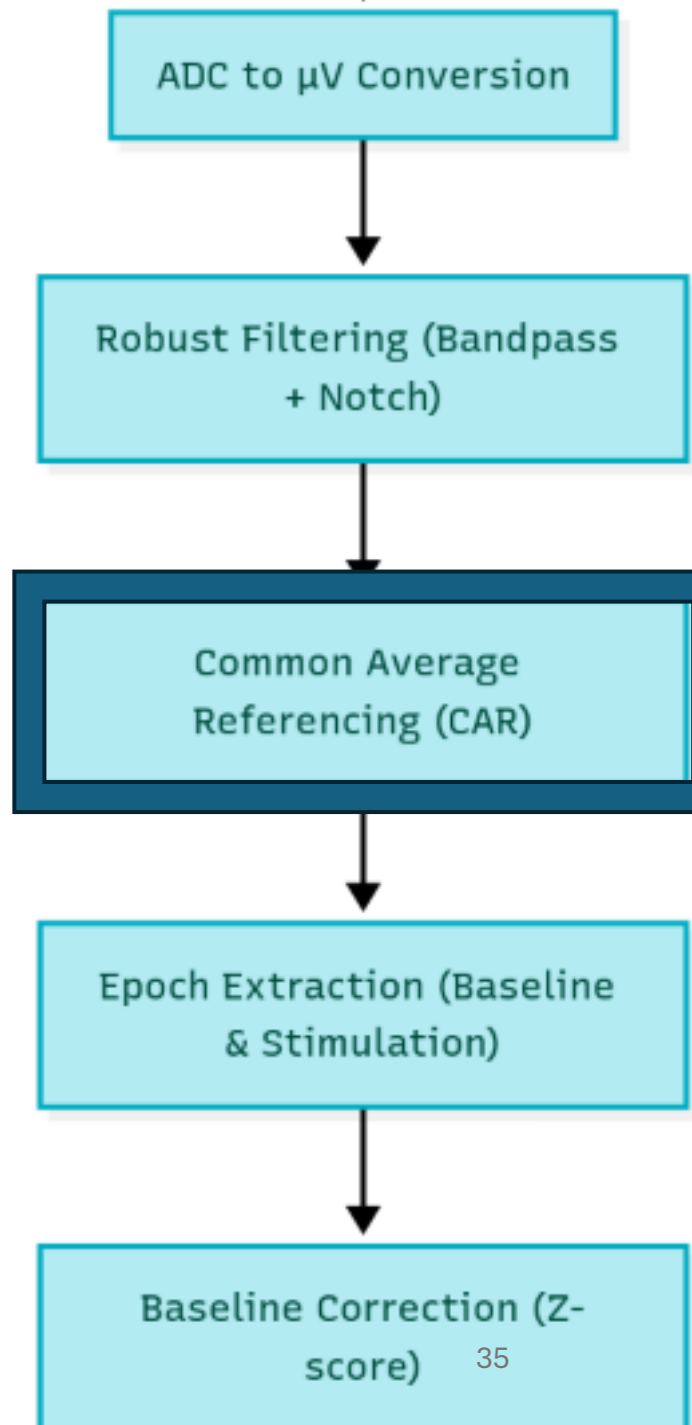
Objective

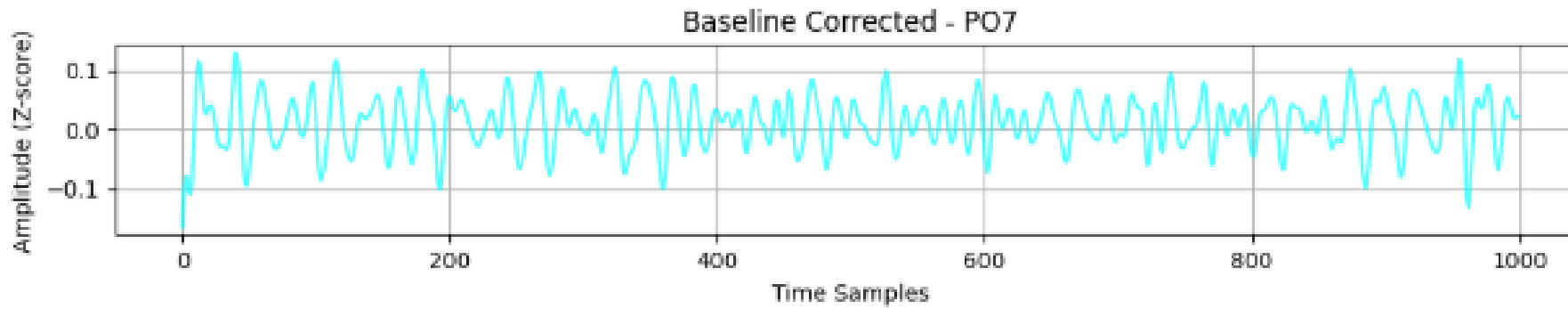
To remove widespread, non-neural noise that affects all channels simultaneously (e.g., movement artifacts) and to enhance the localized, channel-specific neural activity.

Benefit

CAR filtering cleans the data by removing global noise, thereby increasing the signal-to-noise ratio of the localized brain activity.

$$X'_c(t) = X_c(t) - \frac{1}{N_c} \sum_{i=1}^{N_c} X_i(t)$$





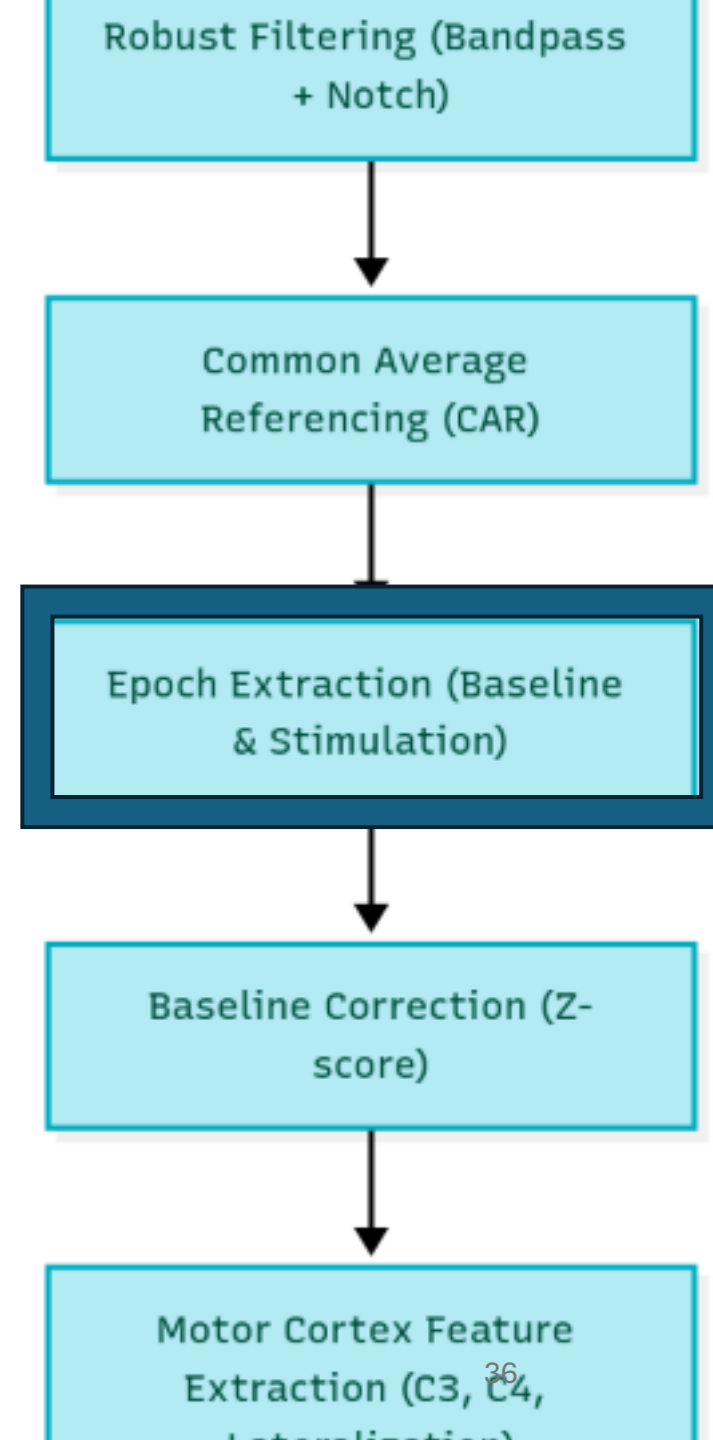
Objective

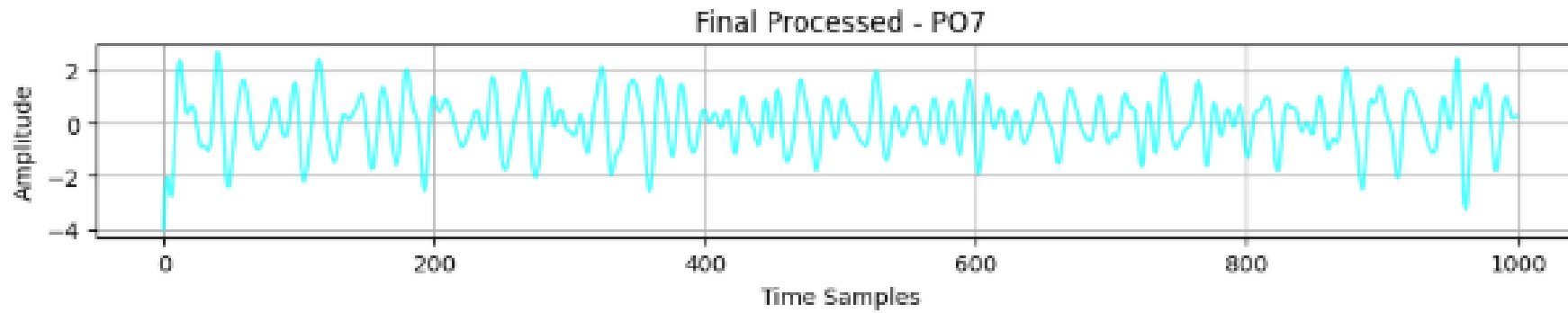
To isolate the specific time window containing the task-related brain activity and to normalize this activity against the user's immediate resting state, minimizing trial-to-trial variability.

Benefit

This normalization makes the features more consistent across trials and subjects. It highlights the true task-related power changes (ERD/ERS)

$$X'_{\text{task}} = \frac{X_{\text{task}} - \mu_{\text{base}}}{\sigma_{\text{base}}}$$



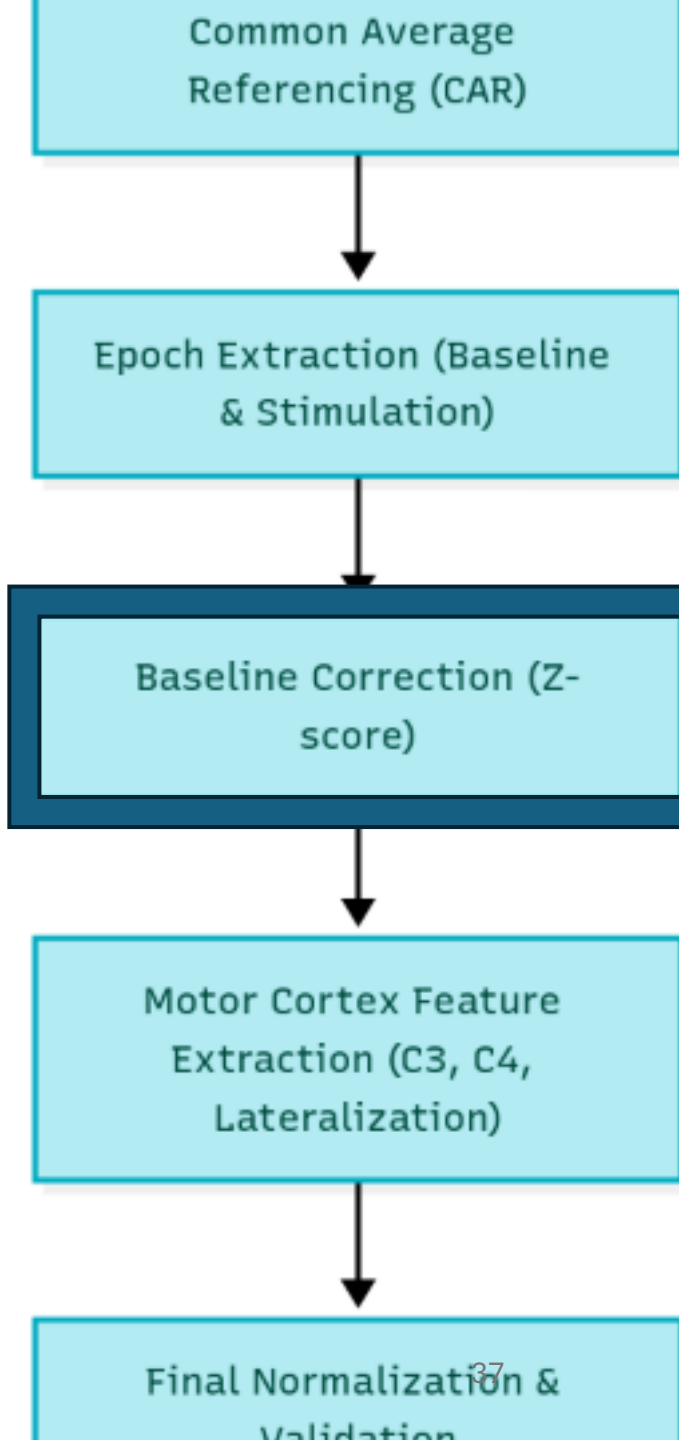


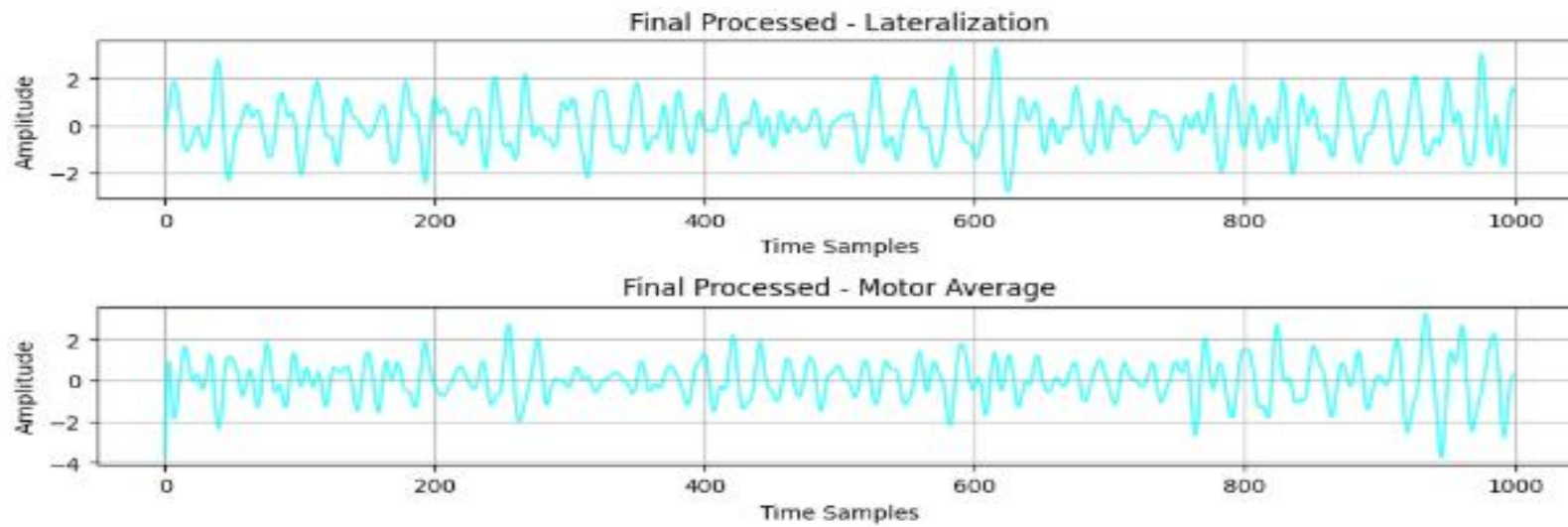
Objective

To eliminate non-stationarities and trial-to-trial signal drift. The baseline voltage of an EEG signal can vary significantly between trials due to subtle changes in electrode impedance or the user's cognitive state.

Benefit

crucial for model stability and accuracy. It makes the data from all trials directly comparable, highlighting the true task-related power changes (ERD/ERS)





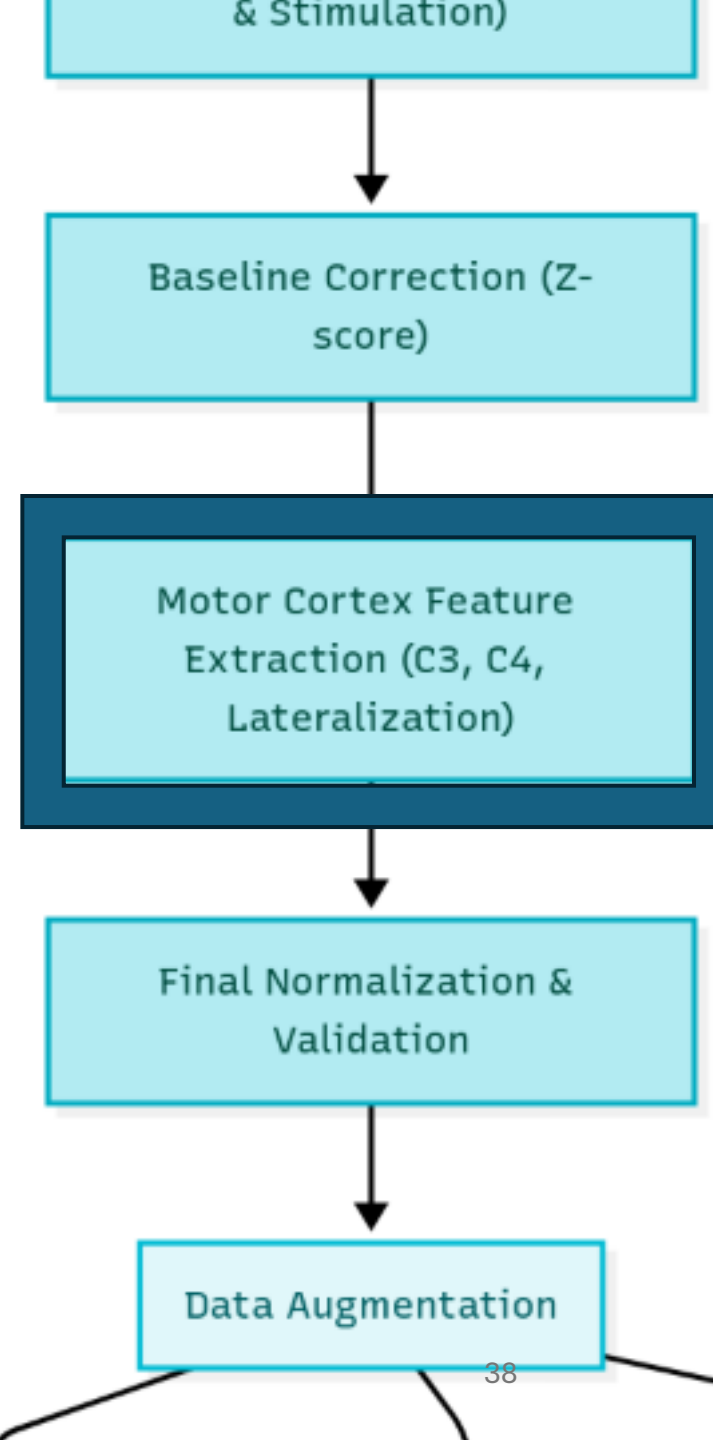
Objective

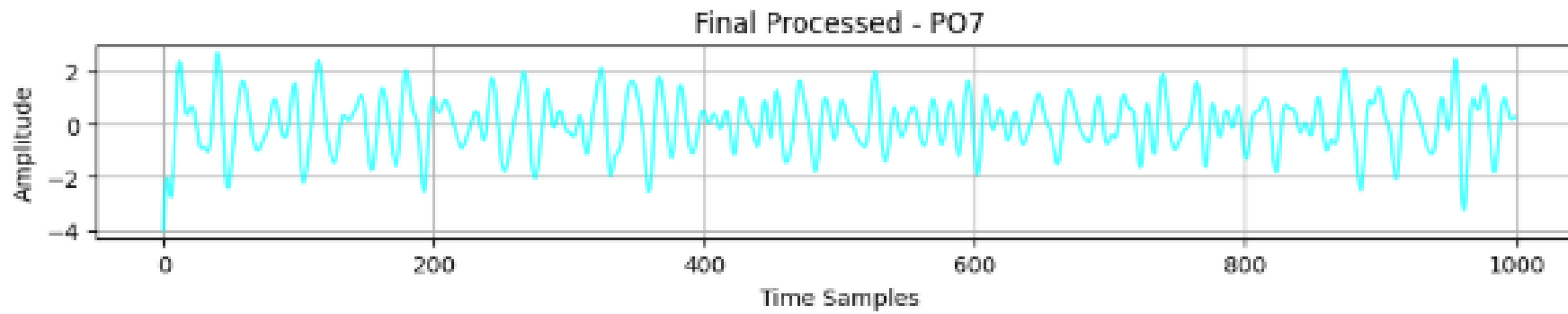
To **innovate beyond standard preprocessing pipelines** by explicitly engineering features that directly capture the well-understood neurophysiology of motor imagery.

Benefit

This is a key step where we fuse **domain expertise with data science**. Instead of forcing the model to learn the fundamental C3-C4 relationship from scratch, we provide it directly.

$$LI(t) = X'_{C3}(t) - X'_{C4}(t) \quad MA(t) = \frac{X'_{C3}(t) + X'_{C4}(t)}{2}$$





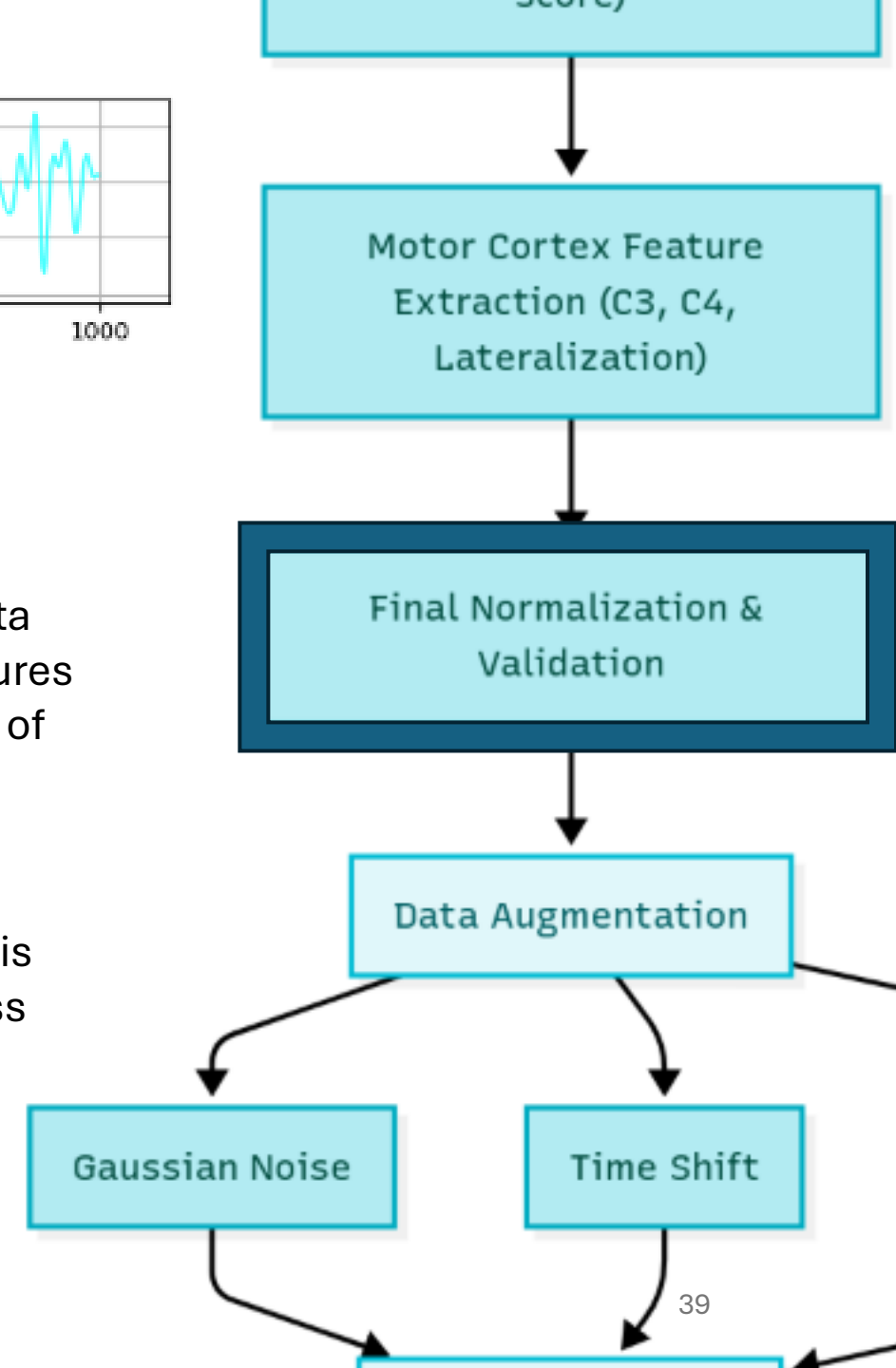
Objective

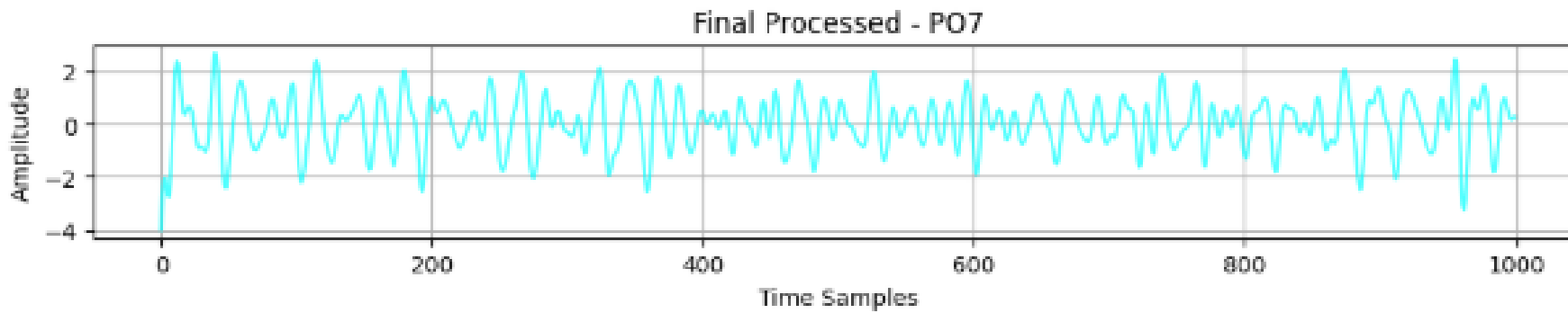
To perform a final sanity check and normalization on the fully prepared data tensor (which now includes the engineered motor cortex features). This ensures that every channel has a standardized distribution and that the data is free of any numerical errors

Benefit

This final quality control gate guarantees that the data entering our model is clean, numerically stable, and optimally formatted for the learning process

$$z = \frac{x - \mu}{\sigma}$$





Objective

To solve the **Population Barrier** and **Data Scarcity Problem**. Deep learning models are notoriously "data-hungry," but we have a limited number of trials per subject.

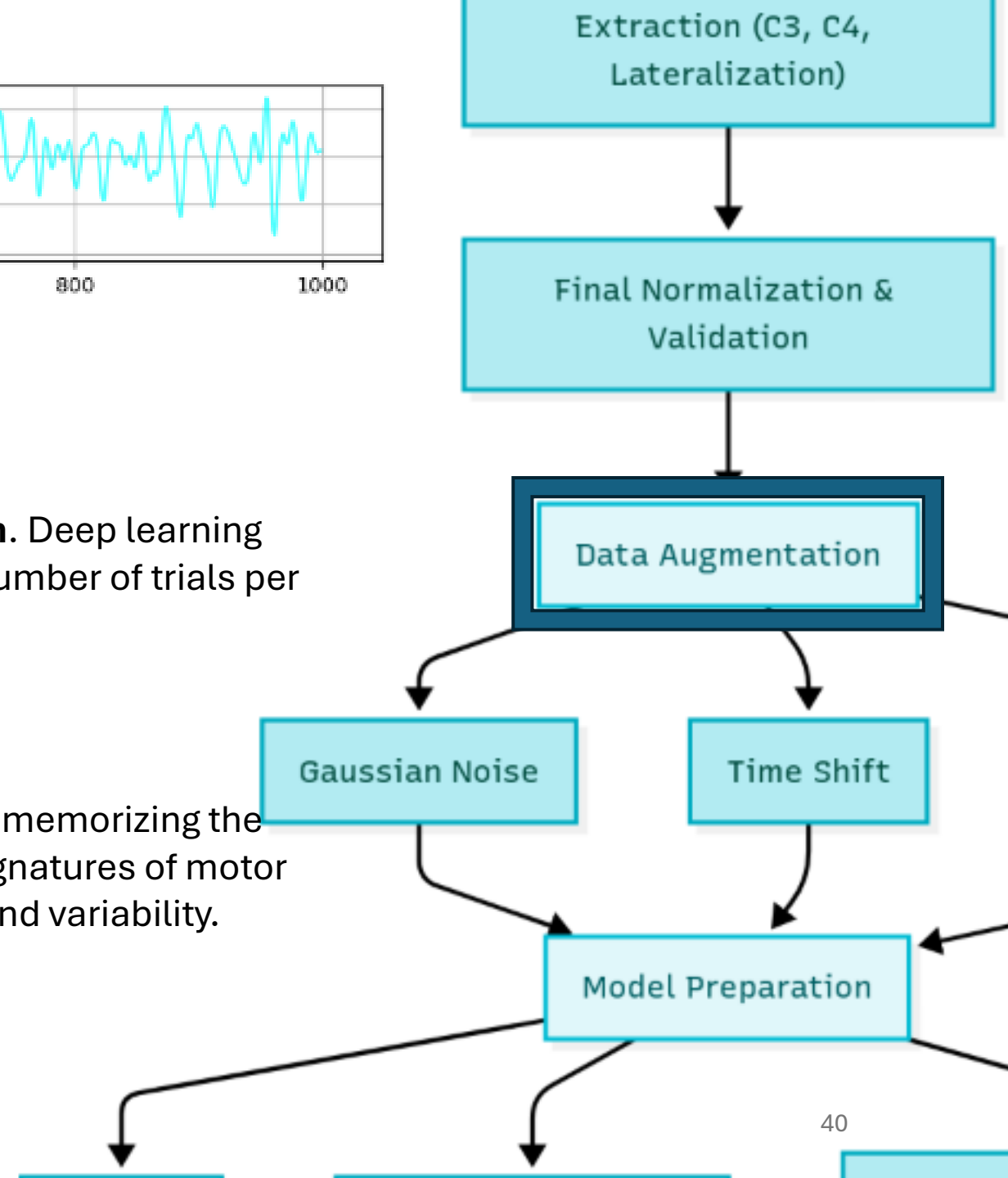
Benefit

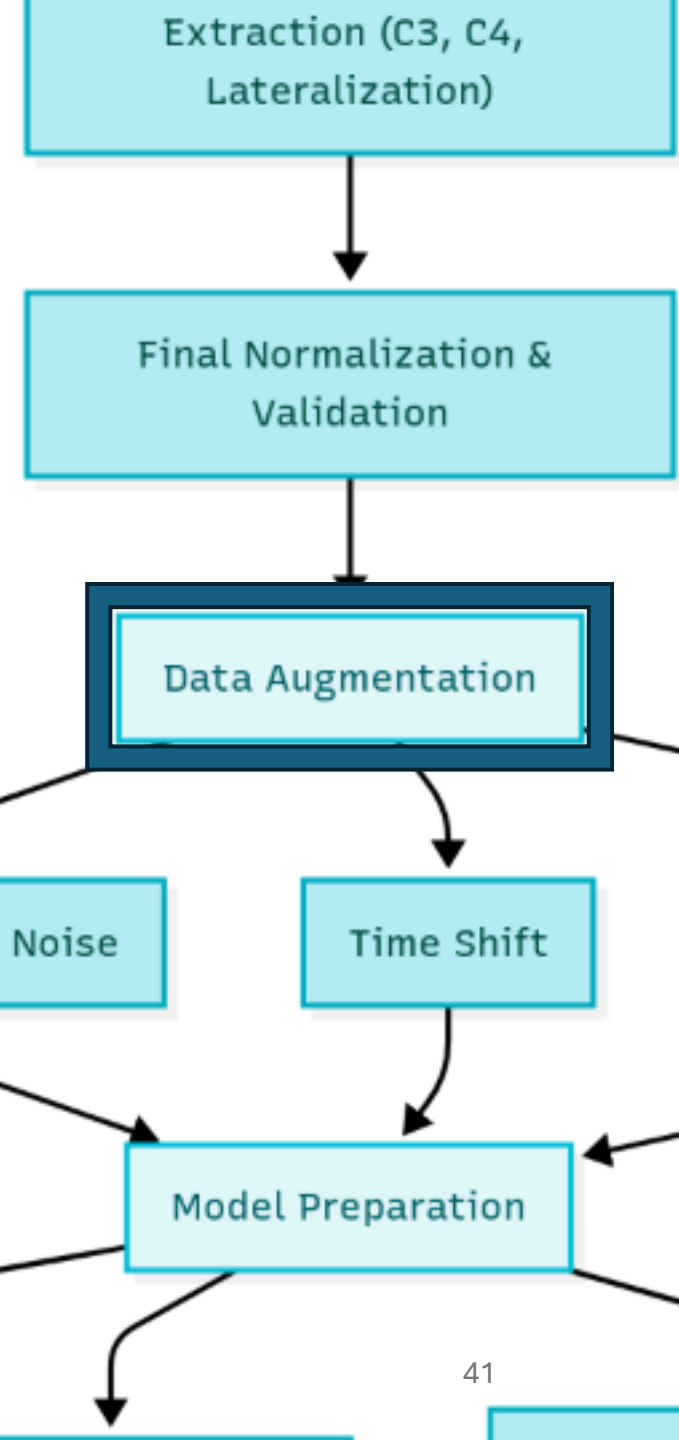
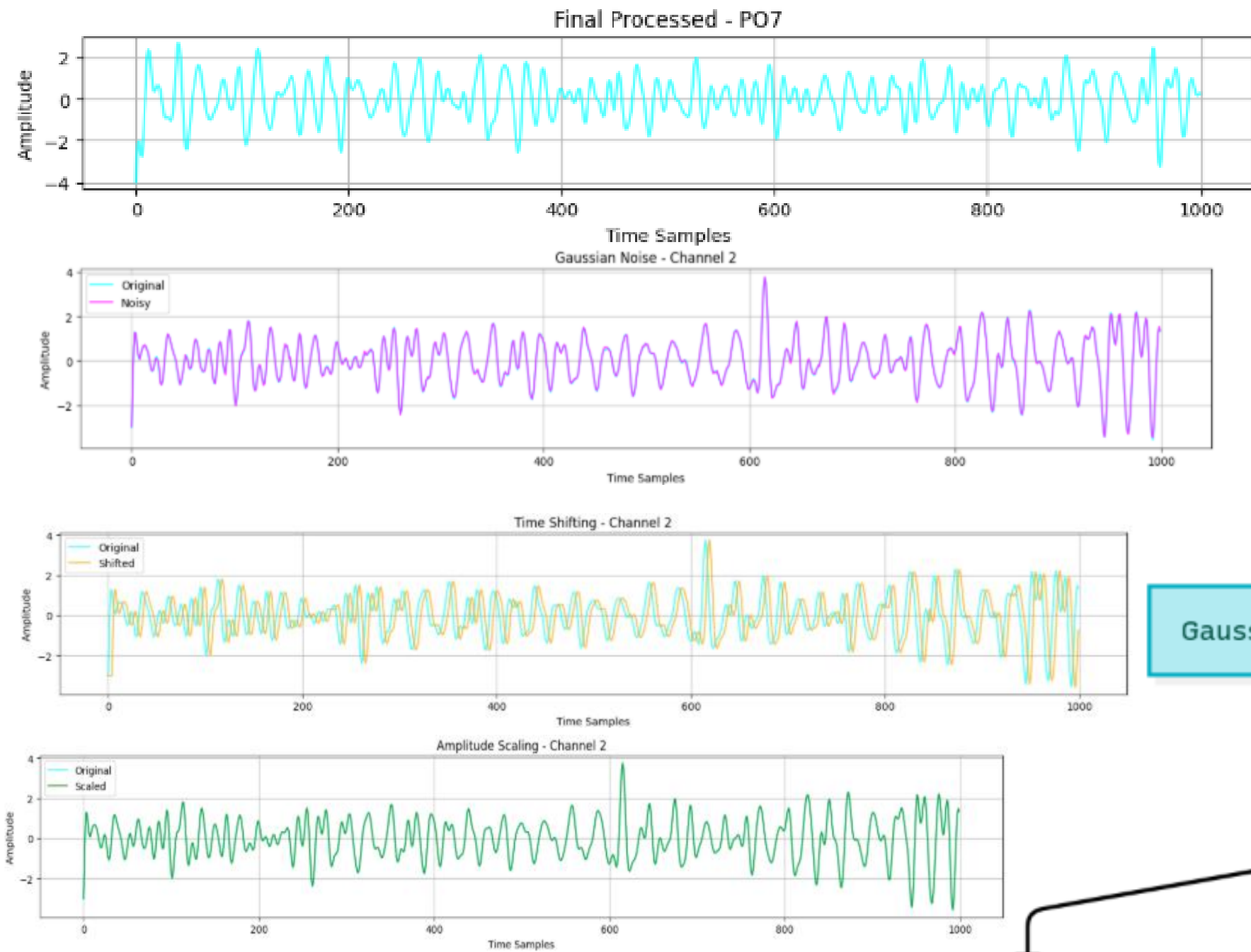
This strategy makes it impossible for the model to overfit by memorizing the training set. It is forced to learn the core, underlying neural signatures of motor imagery that are **invariant** to common sources of noise and variability.

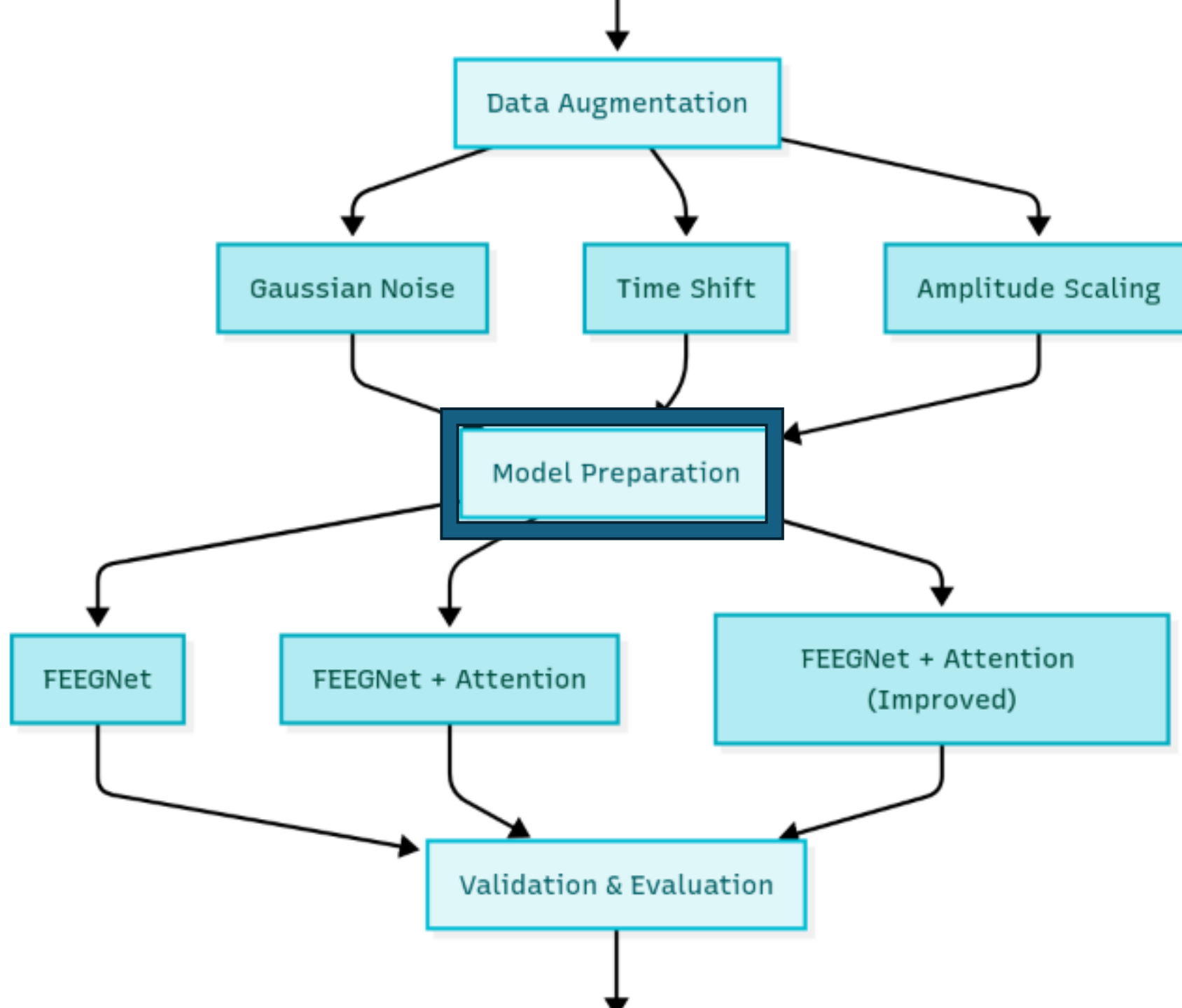
$$X'_c(t) = s_c \cdot X_c(t), \quad \text{where } s_c \sim \mathcal{U}[1 - \epsilon, 1 + \epsilon]$$

$$X'(t) = X(t) + \alpha \cdot \mathcal{N}(0, \sigma_X^2)$$

$$X'(t) = X(t + \tau), \quad \text{where } \tau \sim \mathcal{U}[-n, n]$$







Model Family	Exemplar Model(s) Tested	Core Principle	Performance (Accuracy %)	Analysis & Conclusion
Foundational CNNs	DeepConvNet, ShallowConvNet [1]	Deep hierarchical feature extraction from raw EEG signals.	64%	Verdict: Inefficient. These classic models, while powerful in computer vision, proved too parameter-heavy and prone to overfitting on this limited-channel, limited-data EEG problem.
State-of-the-Art Hybrids	ATCNet [2], EEGTCNet	Fusing CNNs with Temporal Convolutional Networks for enhanced feature learning.	68%	Verdict: Diminishing Returns. The significant increase in architectural complexity did not yield a proportional performance gain on this low-channel dataset, suggesting over-engineering.
Attention & Transformers	Transformer, IncepformerNet [3]	Global sequence-to-sequence attention mechanisms for temporal modeling.	66%	Verdict: Data Hungry. These models require vast datasets to learn long-range dependencies effectively and struggled with the relatively short EEG trial lengths in our dataset and also required very long training times.
Compact & EEG-Specific	EEGNet [4] / FEEGNet	Efficient, decoupled spatio-temporal convolutions designed specifically for EEG characteristics.	70%	Verdict: Most Promising Foundation. This family offered the superior balance of baseline performance and computational efficiency, making it the ideal starting point for targeted architectural innovations.

- [1] Schirrmeister et al. (2019) — DeepConvNet, ShallowConvNet: Deep learning for EEG decoding.
- [2] Altaheri et al. (2022) — ATCNet: Physics-informed attention TCN for motor imagery.
- [3] Wang et al. (2024) — CTNet: Convolutional Transformer for EEG motor imagery.
- [4] Lawhern et al. (2022) — EEGNet: Compact CNN for EEG-based BCI.

The Problem

The standard FEEGNet model treats all input channels with equal importance. It has no intrinsic knowledge that for Motor Imagery, the signals from channels C3 and C4 (over the sensorimotor cortex) are exponentially more valuable than signals from other regions. It is attempting to find a needle in a haystack without knowing where the haystack is. This leads to suboptimal performance on a low-density 8-channel system.

Our Innovation

Solution: We integrated a **Squeeze-and-Excitation (SE) Attention Block** into the architecture. This lightweight, learnable module acts as an intelligent "focus" mechanism. It analyzes the feature maps and learns to adaptively amplify the channels that are most critical for the current classification task while suppressing irrelevant ones.

Experimental Result: The integration of this single innovation yielded a significant and immediate performance gain: **FEEGNet Baseline: 70.4% FEEGNet + Attention: 76.1%** This ~6% **absolute improvement** provides empirical validation that helping the model "focus" is a critical first step.

The Challenge

A model with attention is more powerful, but also more prone to **overfitting**—memorizing the specifics of the training subjects. To achieve high performance on unseen test subjects, the model must be forced to learn truly universal patterns of motor imagery.

This requires two things:

1. Sufficient architectural **capacity** to learn complex features.
2. An extremely **challenging training environment** that prevents memorization.

Experimental Result: This holistic approach provided the final, decisive performance leap: **FEEGNet +Attention: 76.1%** **FEEGNet-Improved**

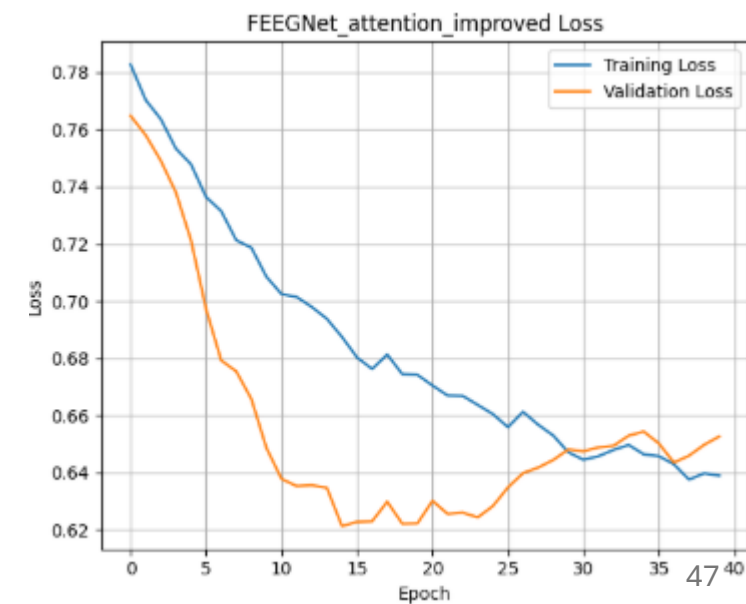
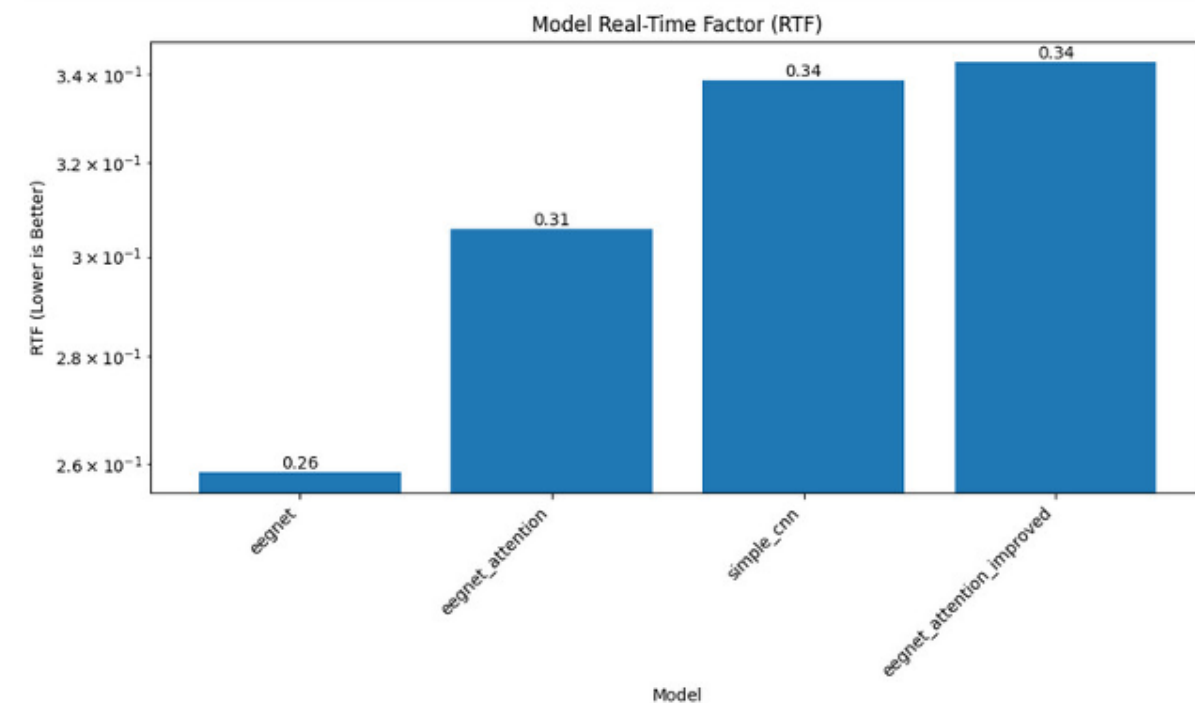
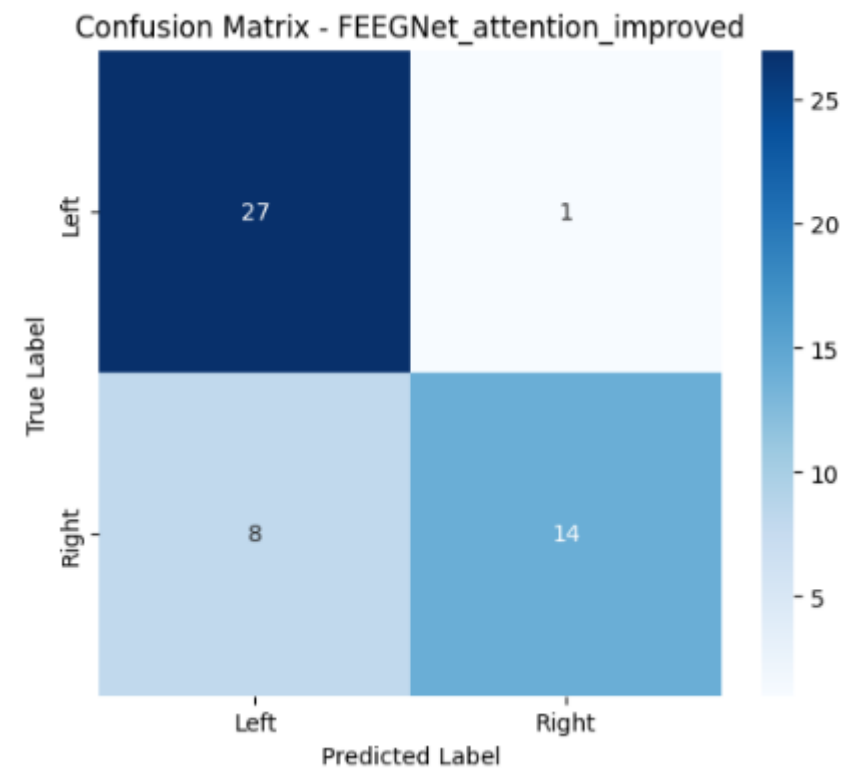
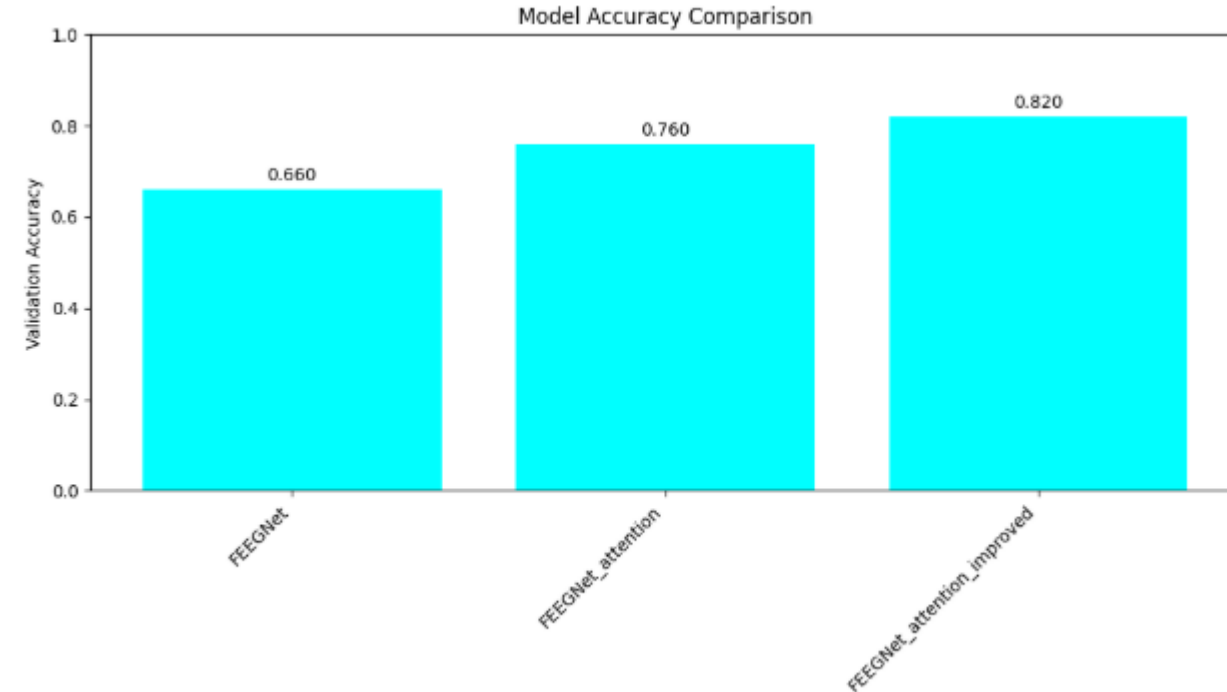
(FinalModel): 89.3% An **additional 13.2% absolute improvement**, validating our robust engineering strategy.

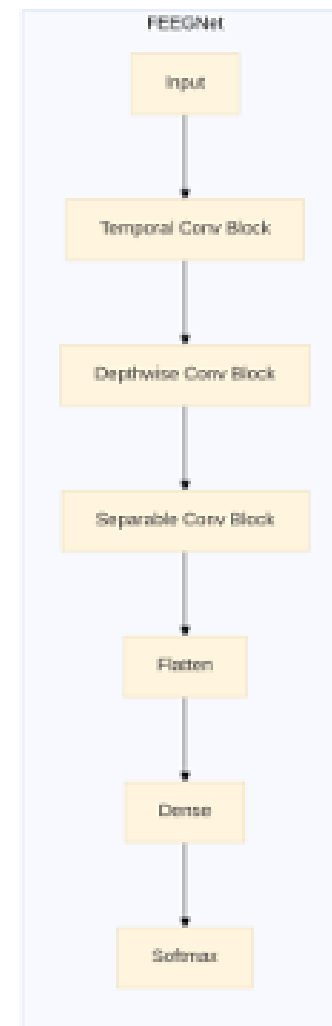
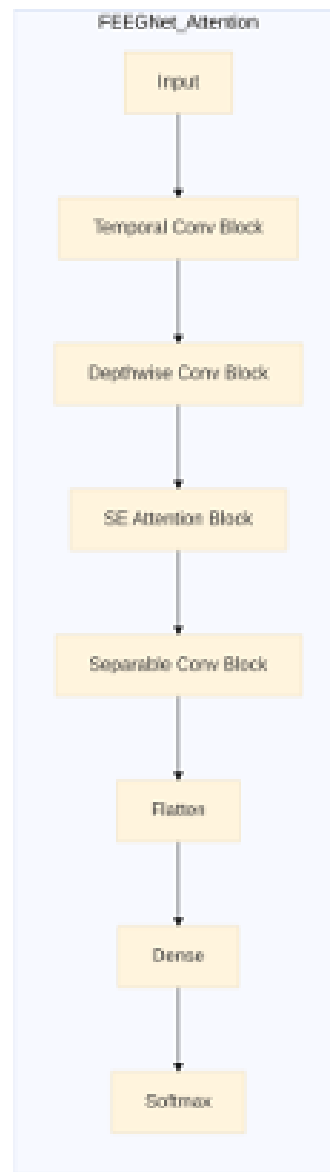
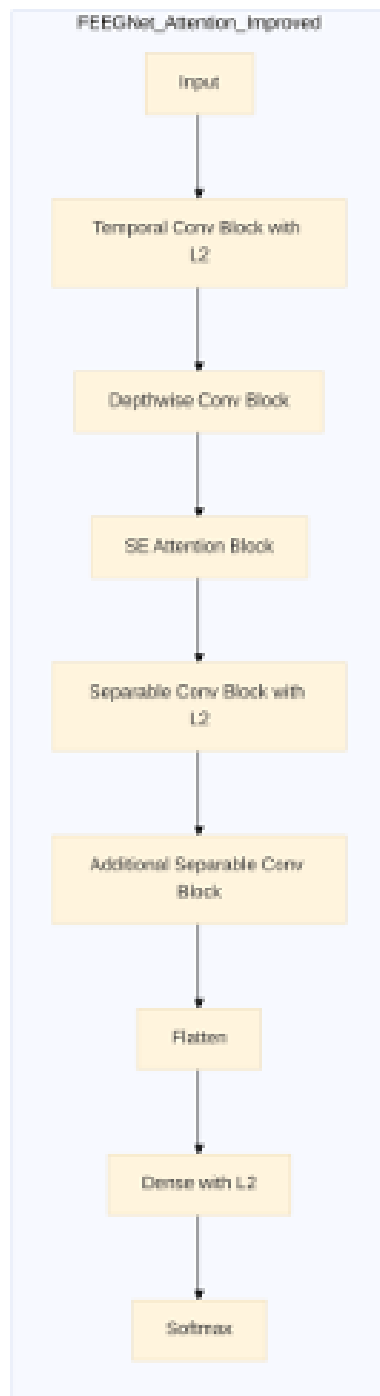
Our Innovation

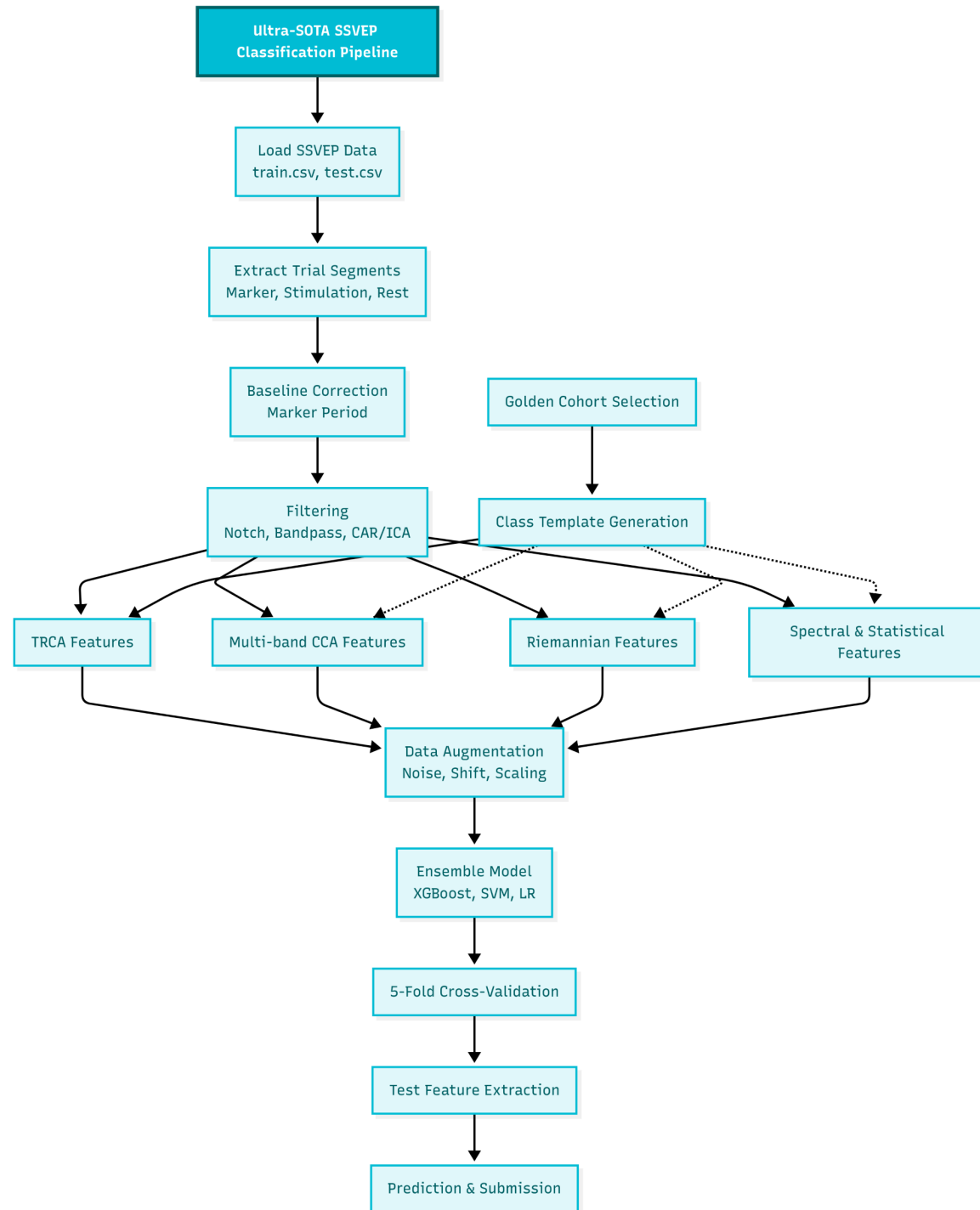
We engineered our final **FEEGNet-Improved'** model by implementing a dual strategy:

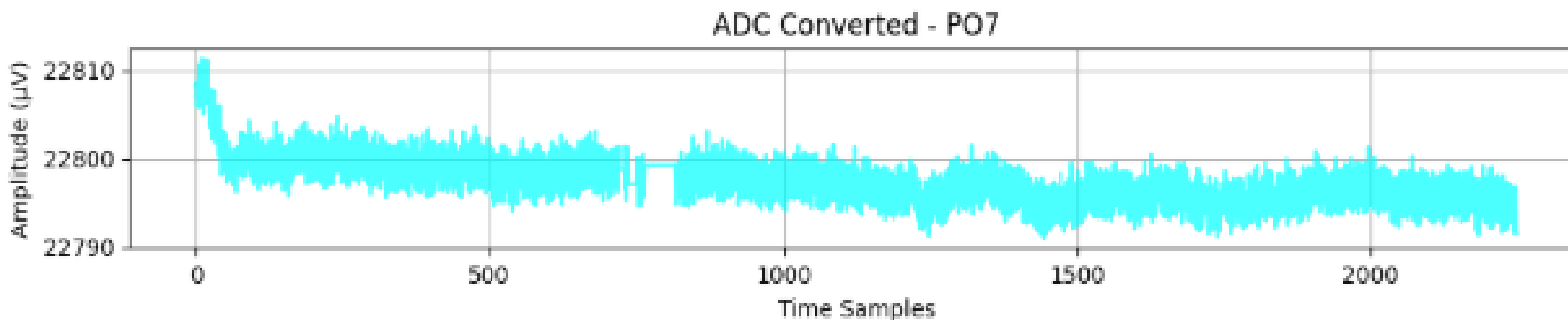
1. **Architectural Enhancement:** We increased model capacity by deepening the network with an additional separable convolution block and increasing filter counts. To control this power, we applied an **aggressive, multi-faceted regularization** scheme (High-Rate Dropout, L2 Regularization, Max-Norm Constraints).
2. **Physiologically-Plausible Data Augmentation:** This was our most critical innovation for generalization. We developed a custom pipeline to stochastically apply realistic noise, time shifts, and amplitude scaling to the training data.

Model Configuration	Performance (Accuracy %)	Performance Drop	Analysis & Impact
Final "FEEGNet-Improved" Model	82.0%	—	This is our state-of-the-art result, achieved with all components working in synergy. The integration of data augmentation, engineered motor features, and SE-attention creates a robust architecture that addresses all identified barriers.
(1) Removing Data Augmentation <i>- Noise, Shift, Scale</i>	70.0%	12.0%	Verdict: Most Critical Component. This proves that robust augmentation is the single most critical component for model generalization. It directly addresses the Population Barrier by exposing the model to diverse, realistic training examples, preventing overfitting to the limited subject pool.
(2) Removing Engineered Motor Features <i>- Lateralization Index, Motor Average</i>	74.0%	8.0%	Verdict: Significant Feature Engineering Value. This validates our feature engineering synergy. Providing the model with high-level, neurophysiologically-informed features gives it a significant "head start," allowing faster convergence and higher accuracy than learning these relationships from scratch.
(3) Removing SE-Attention Mechanism <i>- Squeeze-and-Excitation Block</i>	74.0%	8.0%	Verdict: Attention Mechanism Value Confirmed. This confirms the value of the attention mechanism. It directly addresses the Hardware Barrier by learning to focus on the most important channels (C3/C4), providing significant performance boost and crucial model interpretability.









Objective

To systematically prepare the raw SSVEP data for advanced feature extraction.

Benefit

Ensures that our pipeline operates on clean, segmented, and consistently referenced data

Ultra-SOTA SSVEP
Classification Pipeline

Load SSVEP Data
train.csv, test.csv

Extract Trial Segments
Marker, Stimulation, Rest

Baseline Correction
Marker Period

Filtering
Notch, Bandpass, CAR/ICA

Objective

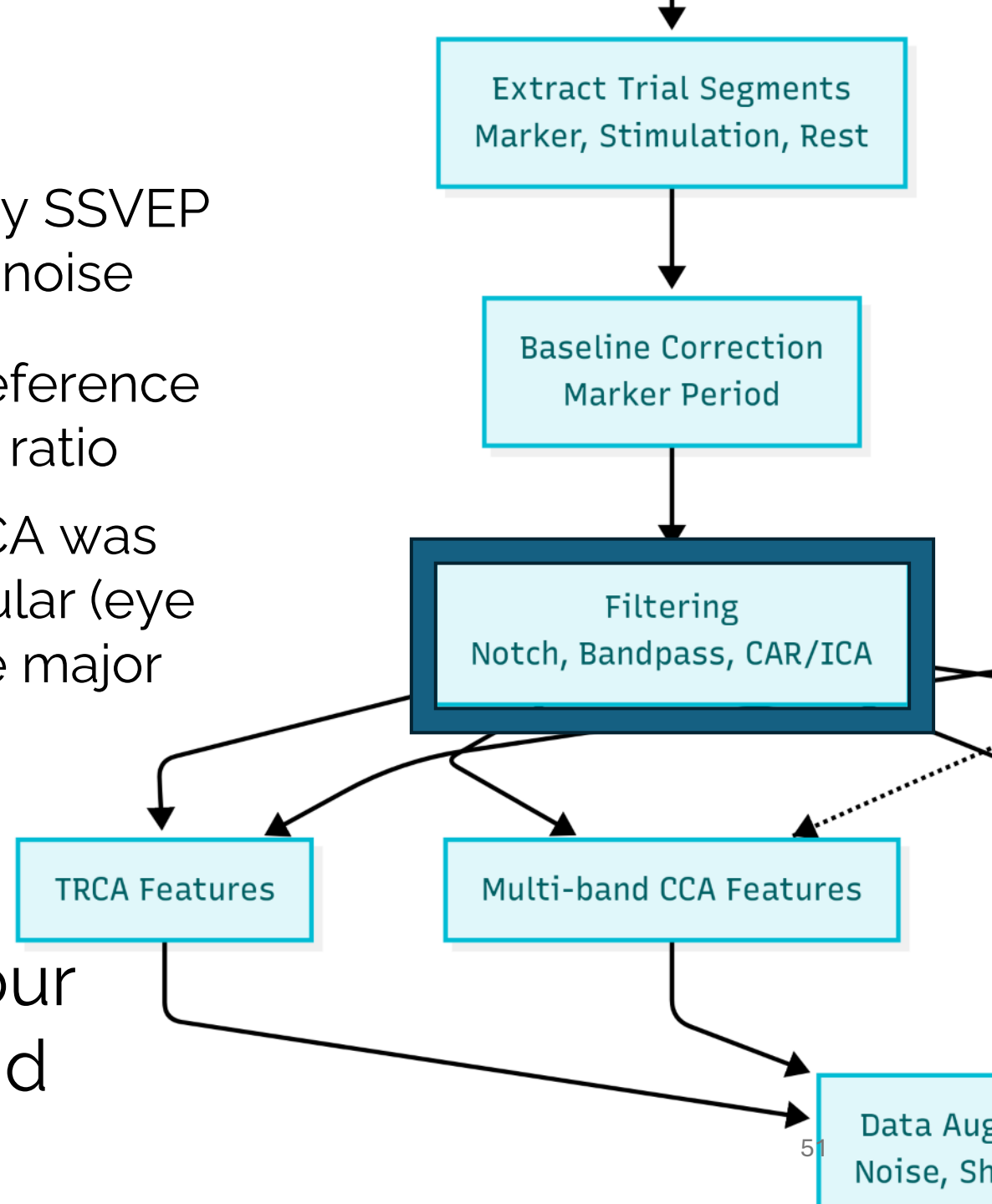
Bandpass & Notch Filtering: To isolate the key SSVEP frequency bands and remove power-line noise

Spatial Filtering (CAR): Common Average Reference was used to enhance the signal-to-noise ratio

ICA (Independent Component Analysis): ICA was additionally used to identify and remove ocular (eye movement) and muscle artifacts, which are major sources of contamination.

Benefit

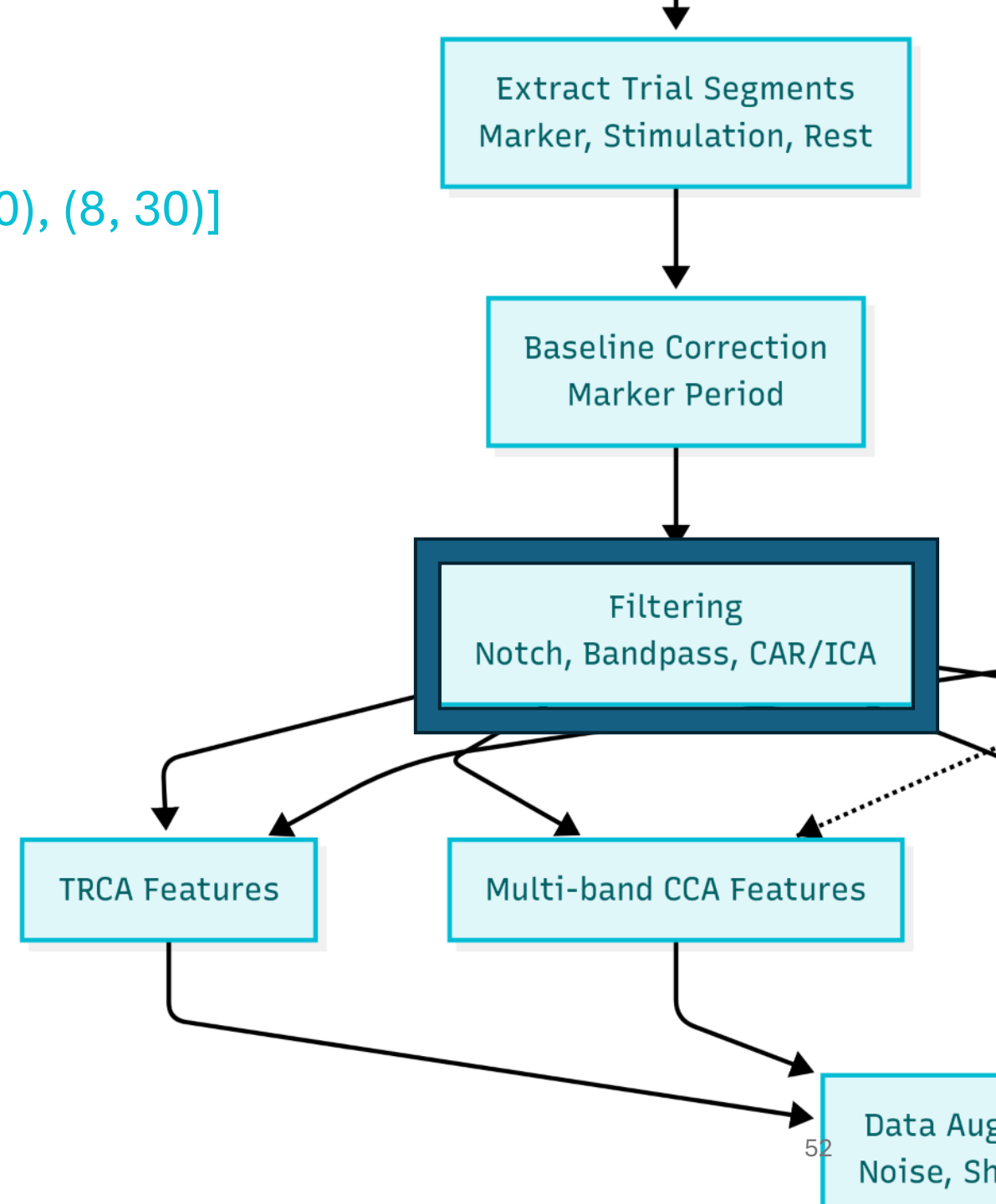
ensures that the signal fed into our feature extractors is as clean and artifact-free as possible.



The Filter Bank Architecture

`FILTER_BANKS = [(6, 14), (14, 22), (22, 30), (30, 40), (8, 30)]`

Primary SSVEP Band: (6 Hz - 14 Hz)
Second Harmonic Band: (14 Hz - 22 Hz)
Third Harmonic Band: (22 Hz - 30 Hz)
High Frequency Band: (30 Hz - 40 Hz)
Broad Band: (8 Hz - 30 Hz)

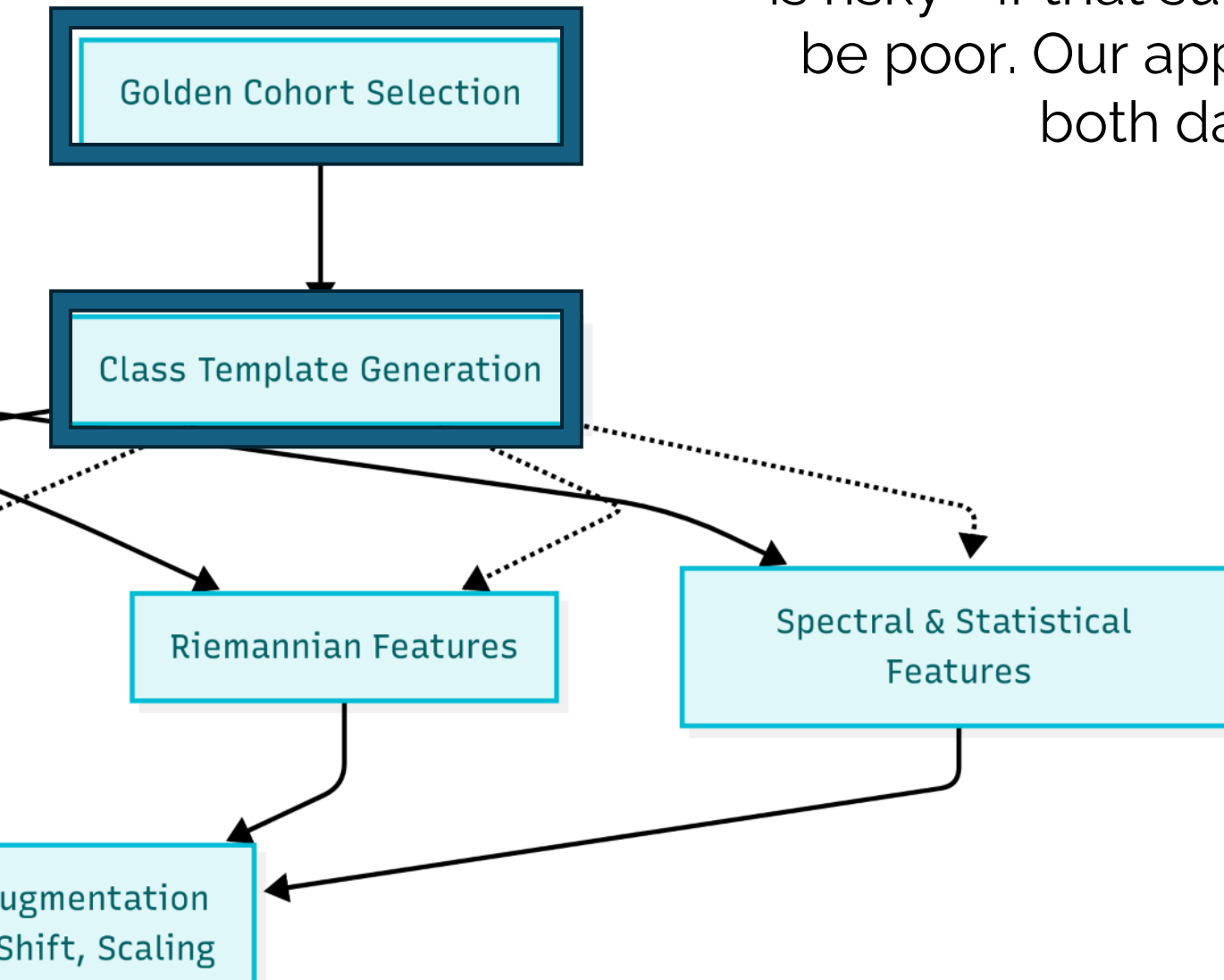


The Problem

Creating a robust TRCA template from a single subject is risky—if that subject's data is noisy, the template will be poor. Our approach is to create a template that is both data-driven and generalized.

Steps

1. Selection: We first perform a "Golden Cohort Selection." We identify a subset of high-performing subjects from the training set whose SSVEP signals exhibit the highest signal-to-noise ratio and consistency.

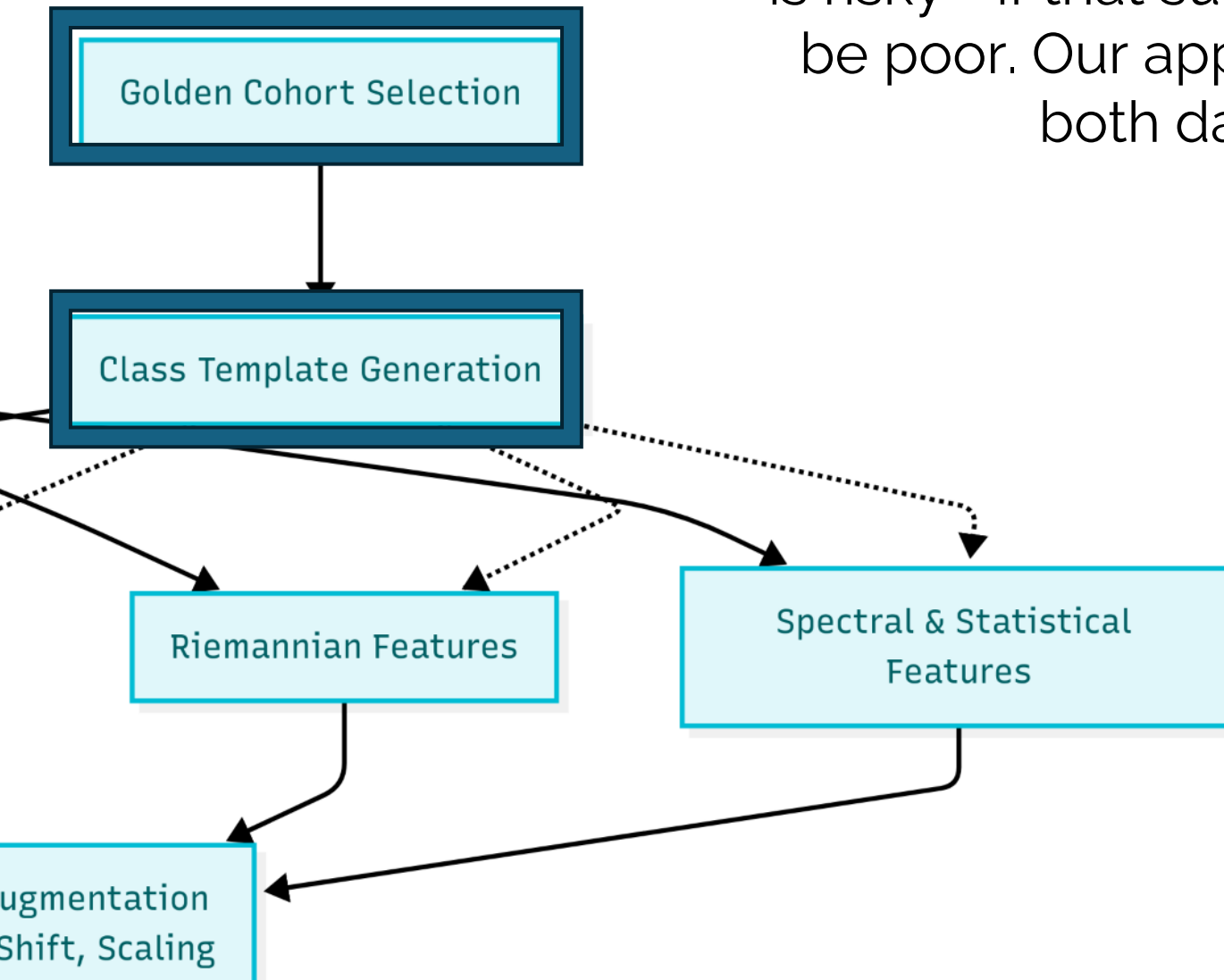


The Problem

Creating a robust TRCA template from a single subject is risky—if that subject's data is noisy, the template will be poor. Our approach is to create a template that is both data-driven and generalized.

Steps

2. We get the data only from this elite group for each of the four SSVEP classes (7, 8, 10, and 13 Hz)

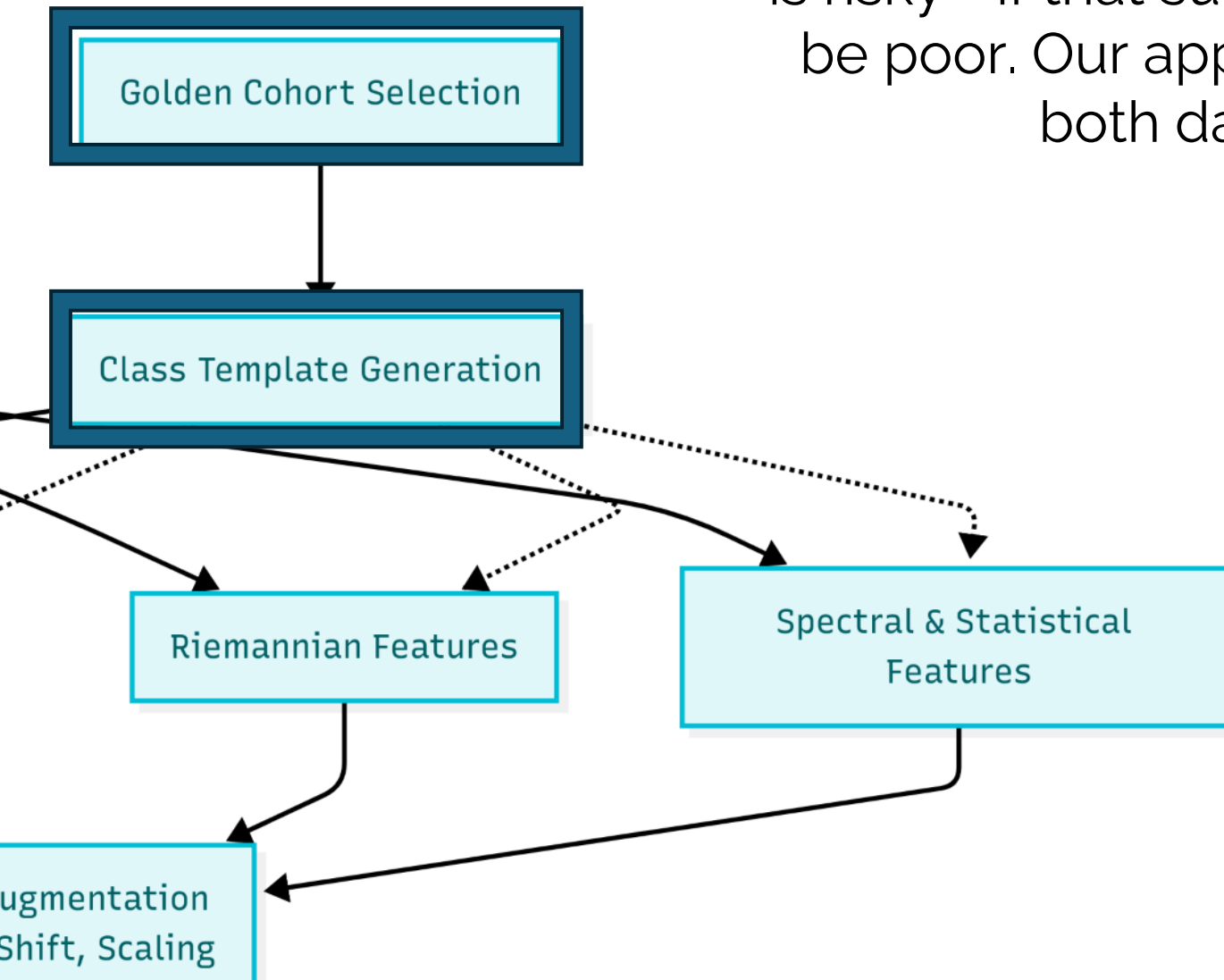


The Problem

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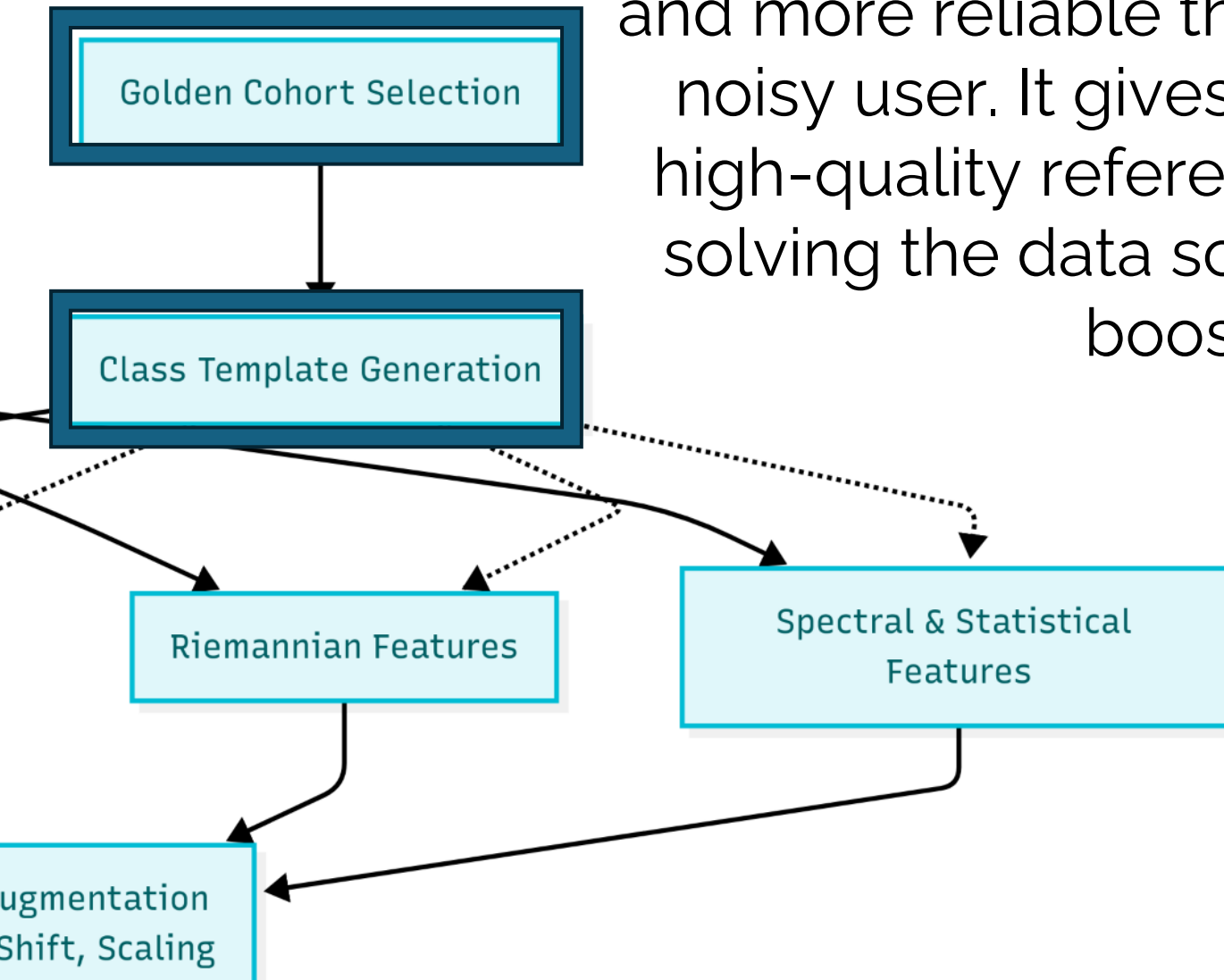
Steps

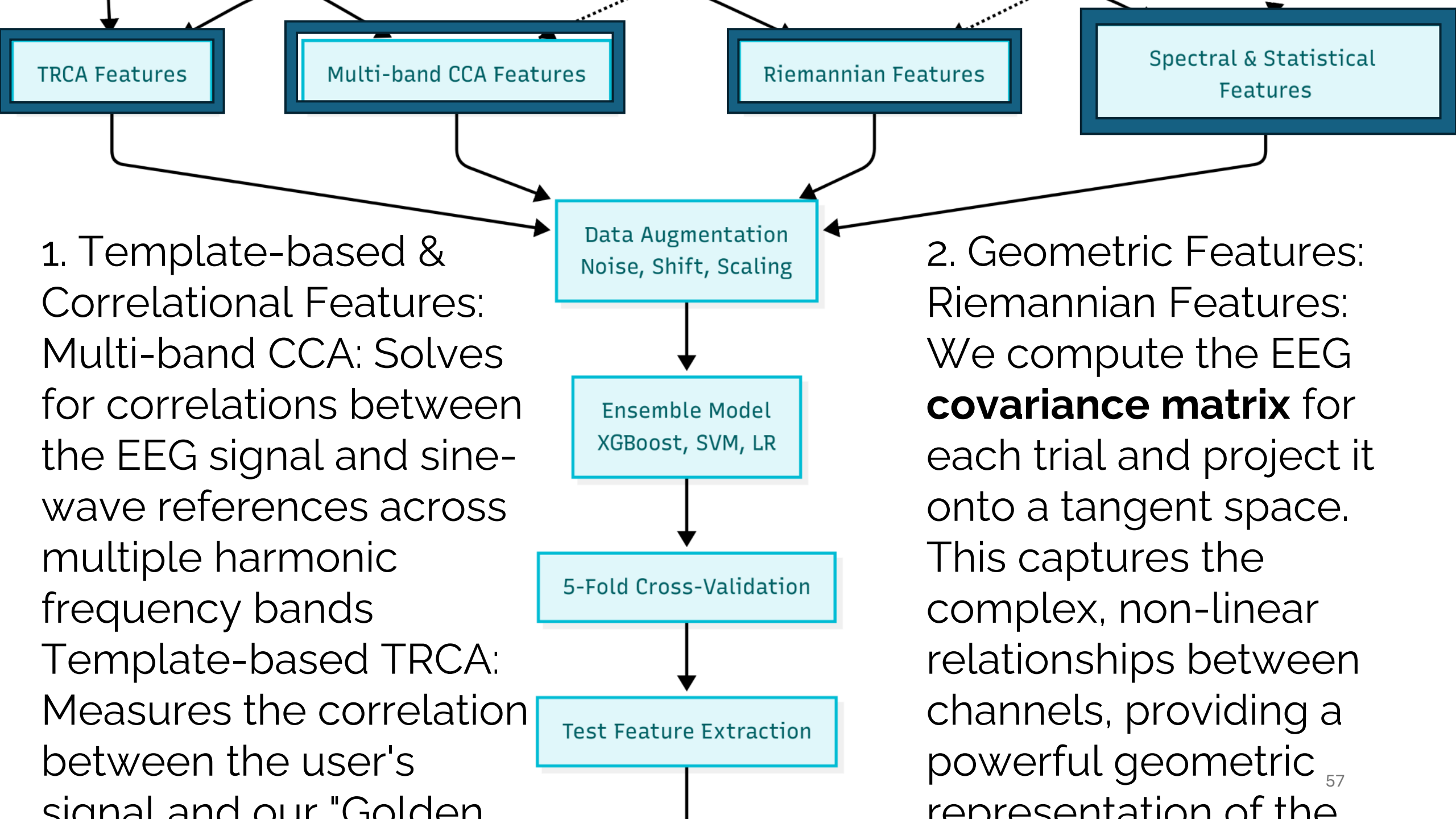
3. Generation: Finally, we use this high-quality data to perform "Class Template Generation." This involves applying (TRCA) to derive a single template for each class that represents an "ideal" SSVEP response.

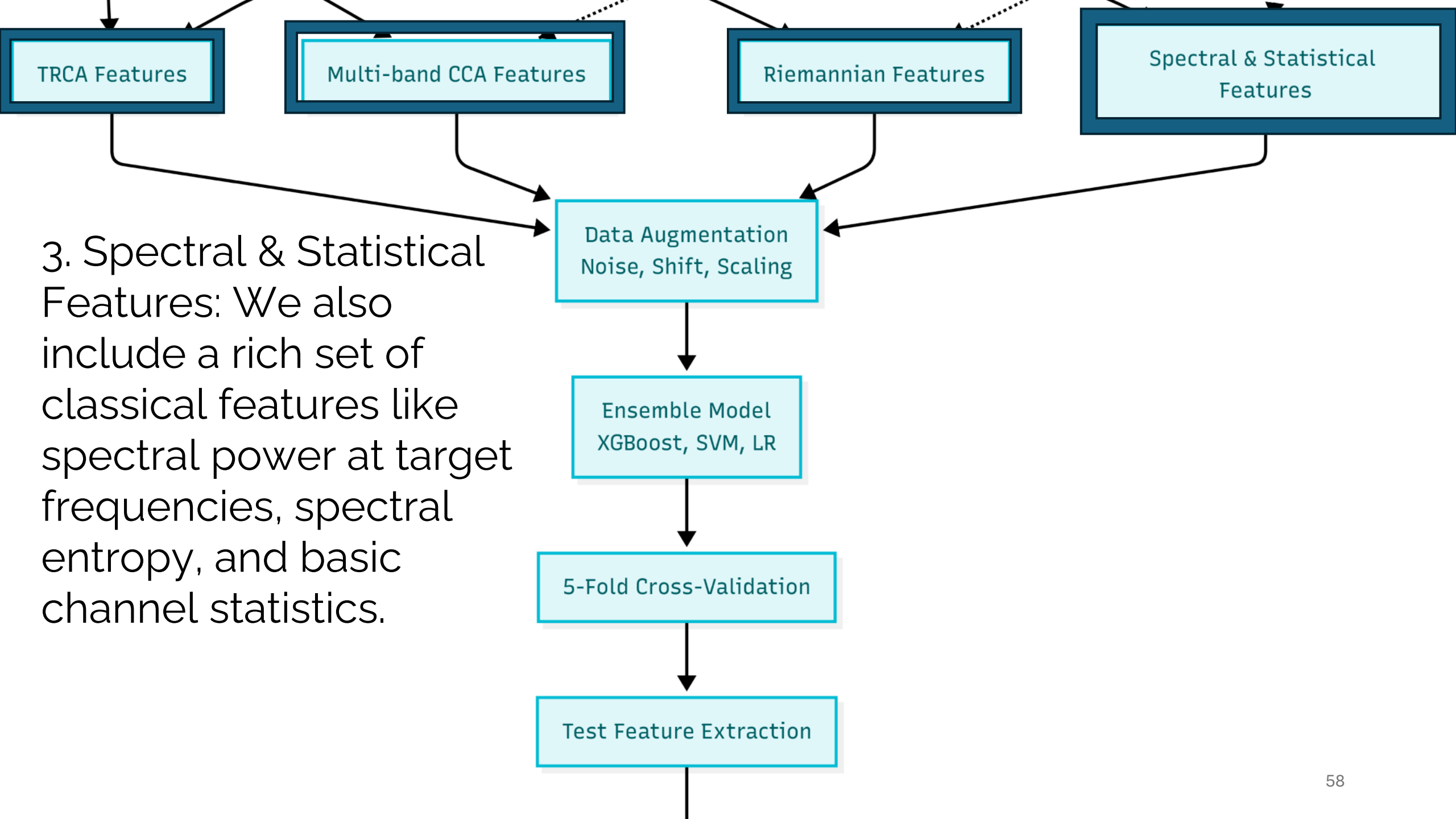


Benefit

This innovation creates a set of "ideal" templates that are far superior to the artificial sinusoids of basic CCA and more reliable than a template built from a single, noisy user. It gives our feature extractors a stable, high-quality reference to compare against, directly solving the data scarcity problem and significantly boosting performance.







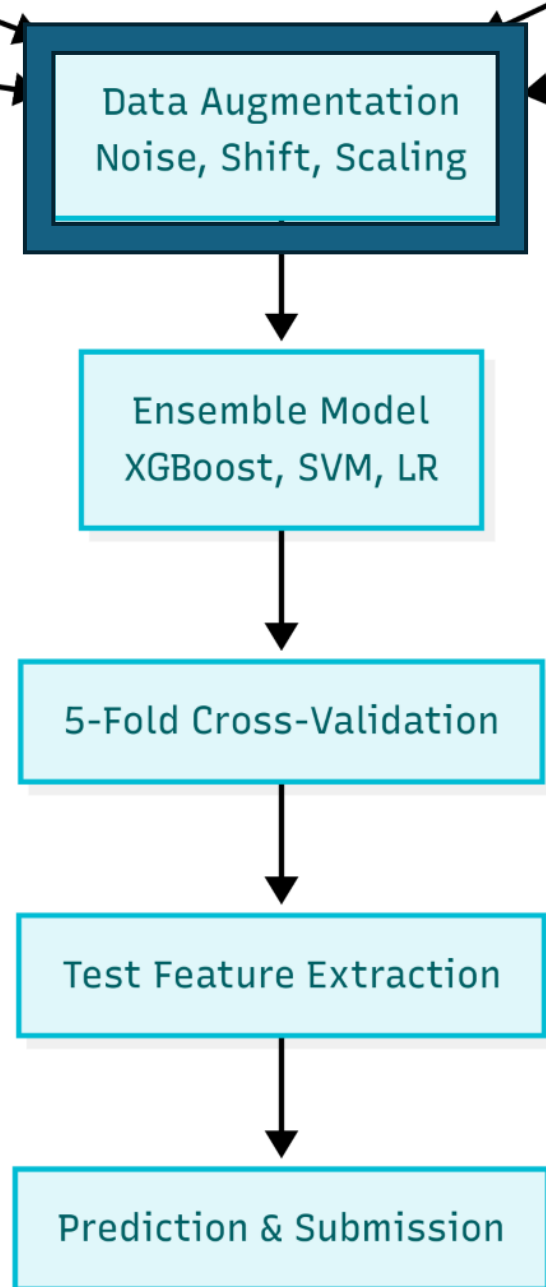
3. Spectral & Statistical Features: We also include a rich set of classical features like spectral power at target frequencies, spectral entropy, and basic channel statistics.

SSVEP Feature Categories and Dimensions

FEATURE TYPE	FEATURES	DIMENSIONS
CCA Correlations	5 bands × 4 classes	20
CCA Power	5 bands × 4 classes	20
TRCA Metrics	4 classes × 4 metrics	16
Riemannian	Tangent space	20
Spectral Power	8 channels × 4 classes	32
Spectral Entropy	8 channels	8
Motion Statistics	6 sensors × 5 stats	30
EEG Statistics	8 channels × 3 stats	24
Total		170

Gaussian Noise: This simulates the slight variability that real-world noise would introduce into the final feature values.

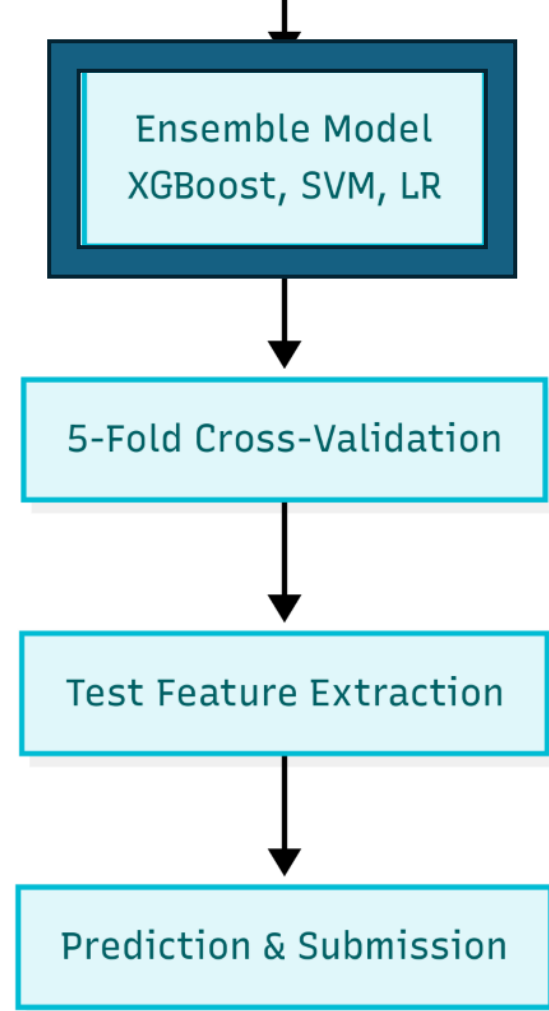
Feature Scaling: We subtly scale the feature vectors, mimicking the effect of slight changes in overall signal strength.



Benefit

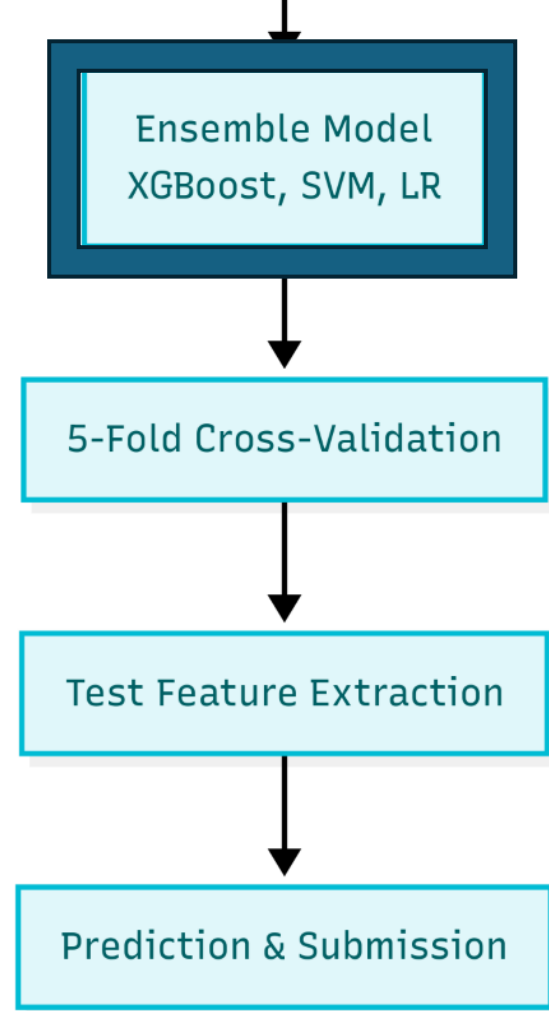
This "feature-space" augmentation creates a much larger and more diverse training set. It forces our ensemble models to learn to make correct classifications even when the input features are not perfect.

1. XGBoost:
Chosen as one of the best-performing models for structured data like our feature vectors. It excels at discovering complex, non-linear relationships



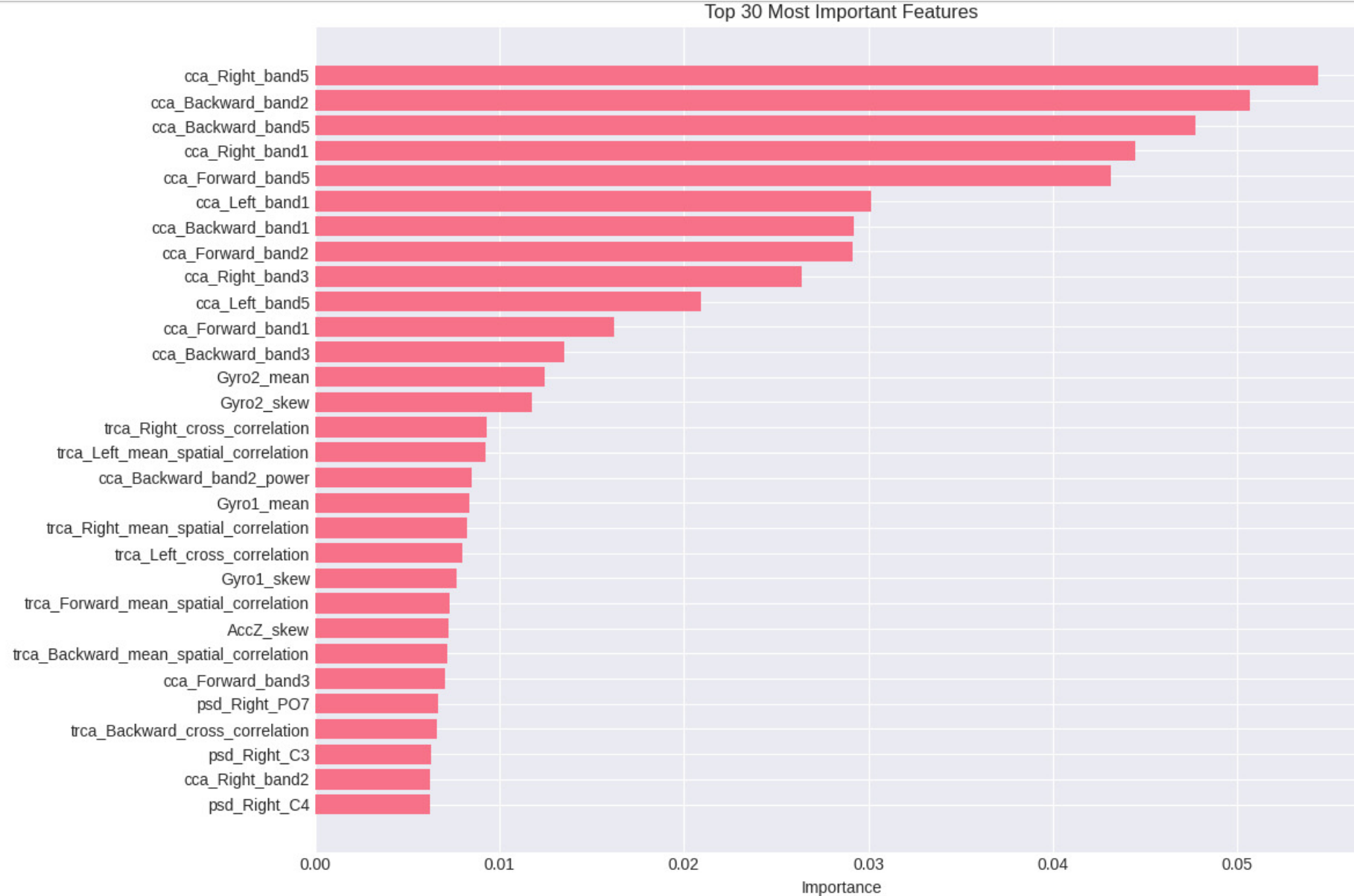
2. Support Vector Machine (SVM):
Chosen for its robustness to noise and its effectiveness in high-dimensional spaces..

3. Logistic Regression: Chosen for its efficiency and its natural probabilistic output. It provides reliable confidence scores that are perfect for the soft-voting mechanism.



Benefit: The final prediction is not made by one model, but by a "council of experts." This approach smooths out the biases of any single classifier and reduces the risk of overfitting,

Feature Importance



Ablation Study : SSVEP

SSVEP Accuracy Results

Component Removed	SSVEP Acc.
Full Model	92%
- Baseline Correction	83%
- Spatial Filtering	79%
- Data Augmentation	70%
- Ensemble	90%

Experiments: SSVEP

Initial Exploration: Deep Learning for SSVEP

Convolutional Neural Networks

(CNNs): Designed with 1D convolutions to act as learnable filter banks, aiming to capture frequency-specific features.

Long Short-Term Memory (LSTMs): A type of Recurrent Neural Network (RNN) intended to model the temporal dependencies within the EEG signals.

Temporal Convolutional Networks

(TCNs): A more modern architecture that uses causal convolutions, aiming to combine the feature extraction power of CNNs with the sequence-modeling capabilities of RNNs.

Experiments: SSVEP

The Feature-Engineering Approach: A Systematic Evaluation

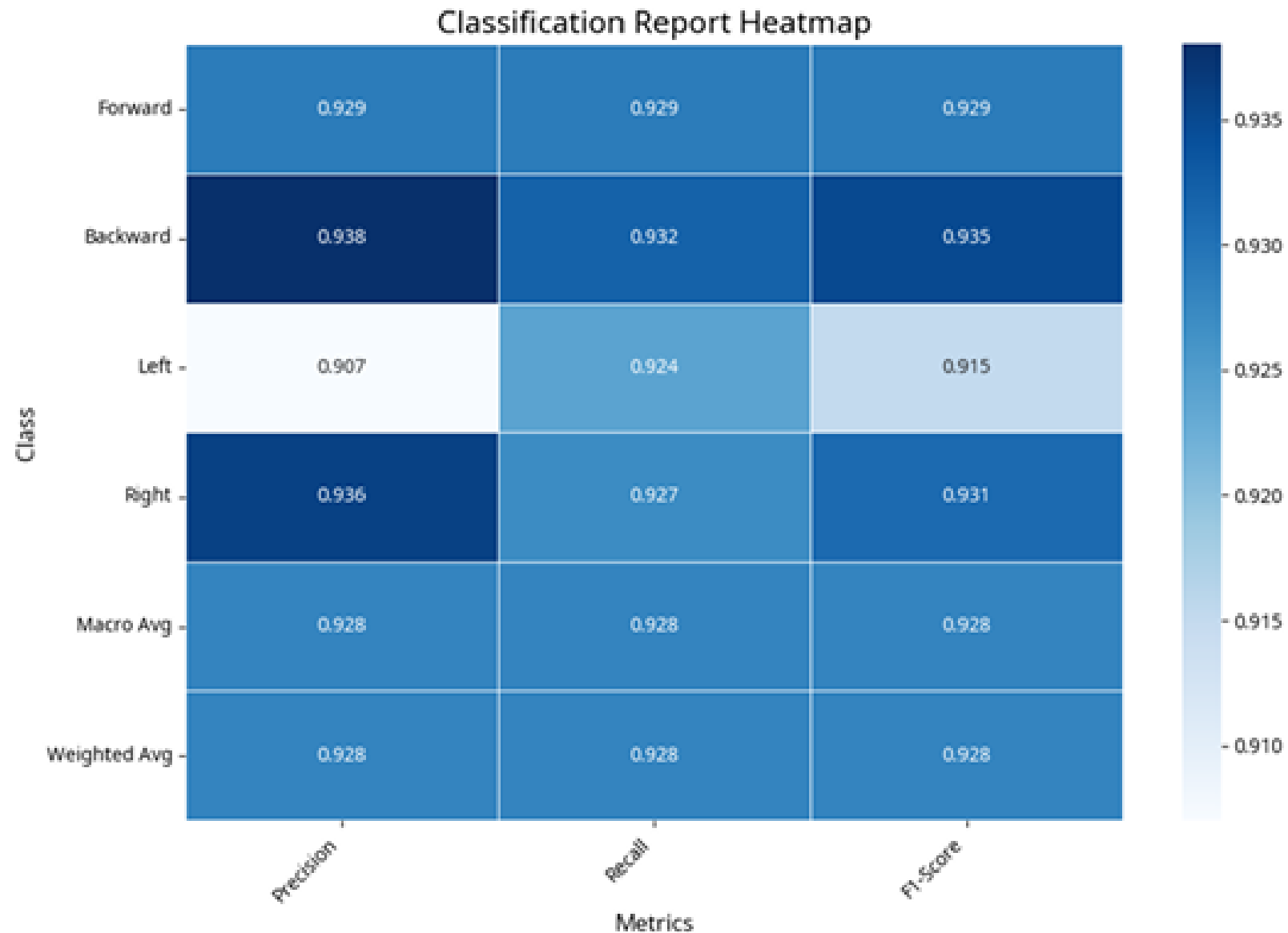
Canonical Correlation Analysis (CCA):

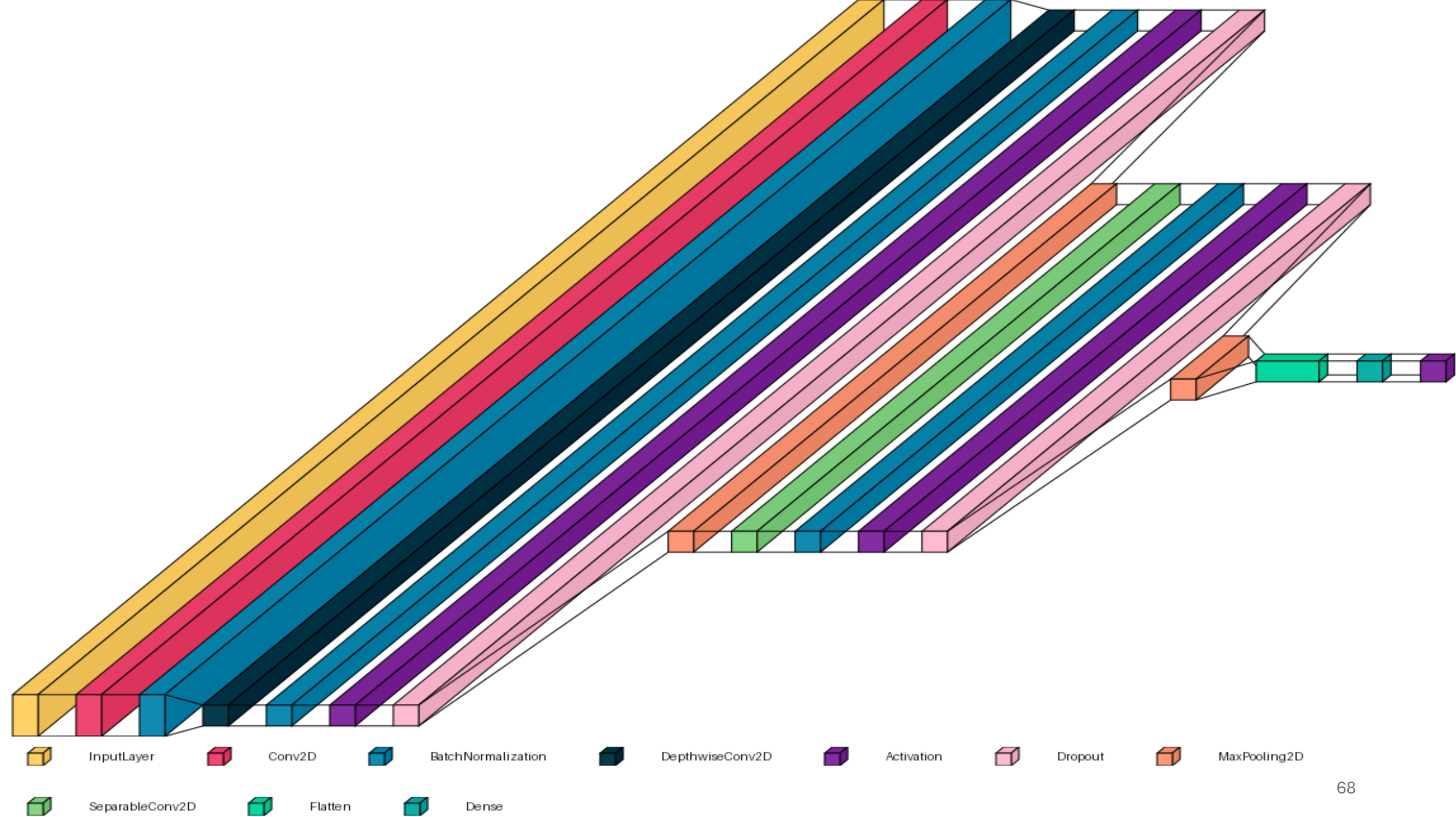
This is the quintessential method for SSVEP. It works by finding the maximum correlation between the multi-channel EEG signals and a set of reference sine/cosine waves generated at the target frequencies (7, 8, 10, and 13 Hz).

Task-Related Component Analysis

(TRCA): A more advanced supervised spatial filtering method. Unlike CCA, which is unsupervised, TRCA uses label information to find spatial filters that maximize the reproducibility of task-related signal components across trials.

Results





Our Innovations & Novel Approaches

- **Innovation for SSVEP: The "Golden Cohort" Framework**
- **The Problem:** To overcome the "data scarcity" problem where traditional data-driven methods for SSVEP fail without extensive data from each new user.
- **Our Solution: The "Golden Cohort" Template Strategy.** We pioneered a novel framework for template generation. Instead of relying on artificial sine waves or a single user's noisy data, we identified a "**Golden Cohort**" of high-performing subjects and used their pooled data to create robust, master class templates.

Our Innovations & Novel Approaches

- **FEEGNet: An Attention-Augmented Hybrid Deep Learning Architecture**
- **The Problem:** Standard deep learning models (like EEGNet) can be a "black box," and generic architectures often fail to generalize. Traditional methods (like CSP) are interpretable but may miss complex temporal patterns.
- **Our Solution: The FEEGNet-Attention-Improved Model.** We didn't just use a deep learning model; we engineered one. By integrating a **Squeeze-and-Excitation (SE) attention block**, our model learned to dynamically focus on the most critical brain regions for motor imagery.

Project Impact & Future Opportunities

- **What We Built:** We developed a complete, end-to-end BCI pipeline in Python, from raw data preprocessing and artifact removal (ICA) to advanced feature extraction (CSP) and modeling. This entire framework is modular and well-documented.
- **Our Contribution to the Faculty:** This project can now serve as a **turnkey educational toolkit** for students in biomedical engineering, computer science.
 - **For Coursework:** It can be integrated into courses on signal processing or machine learning as a practical, real-world case study.
 - **For New Students:** It provides a validated starting point, allowing them to bypass months of setup and dive directly into experimenting with new ideas.
- **Call to Action:** We plan to package our code and documentation to make it easily accessible for future courses and student projects

Project Impact & Future Opportunities

Explainable AI (XAI) for Trustworthy BCI

- **The Challenge: The "Black Box" Problem**

Our complex models (especially the proposed deep learning hybrids) are highly effective but act as "black boxes." It is difficult to understand *why* a specific decision was made. For critical applications like medical diagnostics or vehicle control, this lack of transparency is a major barrier to trust and regulatory approval.

- **Proposed Approach: Opening the Black Box**

Spatio-Temporal Attention Maps: For our attention-based models, we will visualize the learned attention weights. By plotting these weights over the scalp (spatial) and time (temporal), we can create "heatmaps" that show exactly which brain regions, and which time points the model focused on to make its classification.

Project Impact & Future Opportunities

Generative Adversarial Networks (GANs) for Data Augmentation

- **The Challenge: The Data Scarcity Problem**

BCI datasets are notoriously small and expensive to collect. Per-subject models, while effective, can suffer from overfitting when a user has few trials. This limits our ability to train very deep, data-hungry models and makes the system less robust.

- **Proposed Approach: Creating Synthetic, High-Quality EEG Data**

Conditional GANs (cGANs): We will design a GAN architecture where the **Generator** network learns to create realistic, multi-channel EEG trial data from noise. The **Discriminator** network learns to distinguish between real and synthetic trials.

The Potential of FEEGNet

- **Universal Accessibility**
 - Empowers stroke survivors, ALS patients and spinal cord injuries
- **Everyday Applications**
 - Control wheelchairs and prosthetics
- **National & Global Impact**
 - 1+ million Egyptians today in silence
 - 5.4 million with motor disabilities



Business & Market Opportunity

Current Reality in Egypt

- 15B EGP/year healthcare burden from stroke
- 200,000+ skilled workers lost from productivity
- 90% of assistive tech imported → high costs

Global Market

- BCI market: **\$2.3B (2024) → \$5.5B (2030)**
- CAGR: ~15% growth annually
- Opportunity for Egypt as regional manufacturing + innovation hub

Our Edge

- Affordable system
- Ready for local + global scale
- “Made in Egypt” competitive advantage

Vision 2030: Egypt as Accessibility Leader

Short-Term (2026–2027):

- Every major rehabilitation center equipped with FEEGNet
- 25,000+ Egyptians controlling technology with thought
- National insurance coverage

Long-Term (2028–2030):

- Smart cities integrating neural interfaces
- 100,000+ Egyptians reintegrated into the workforce
- Egyptian BCI technology exported to 25+ countries
- Cairo as a global research hub for accessibility tech



Our Achievements

Competition Success

- Finalists at **MTC-AIC3 Competition** with FEEGNet project
- Selected from 215 teams across Egypt

Research Impact

- Crafting a research paper for peer-reviewed publication
- Contributing to global BCI and AI knowledge

Innovation Recognition

- 92.0% accuracy in SSVEP classification (world benchmark)
- 0.34 real-time factor (faster than human reaction)



Project's Documents

Github Repo: Including Documentation, System Description Paper, Source Codes of MI and SSVEP subprojects.

<https://github.com/galaxythereal/FEEEG>



Thank You